The Average Patient: Do Simulated Psychiatric Patients Answer Like Real People?

Anonymous ACL submission

Abstract

Language models are used as simulated psychiatric patients to train clinicians and evaluate AI therapists, and are judged by plausibility or total scores. We give a framework for simulated respondents: an information budget bounds how much simulated patients can differ, and an average-patient hypothesis, that a model answers each item at the level typical of the named condition and severity, implies little variation, a full profile of the condition's typical symptoms and totals at or above the top of the stated severity band. We examine these implications at matched totals on item-level data from eight human samples, including a clinical and an instructed-feigning sample. Across 17 models, simulated patients vary 1.5 to 11 times less than real respondents with the same total and on most questionnaires endorse nearly every symptom at a middling level; on the PHQ-9 their symptoms covary less like real ones. People, honest or feigning, answer "not at all" about as often as genuine patients with the same totals, and more often than every model. Single answers are highly concentrated, and prompt changes, published personas, larger models and reasoning reduce the pattern without removing it. In interviews, patients told a severity are rated as having nearly every symptom, so an assessment agent's recognition of absent symptoms goes untested, and scored against the stated severity the agent places a quarter of them in their band, against about half of patients given a real symptom profile. Having the model word such a profile restores half to two thirds of the variation. Our code is available here.

1 Introduction

Language-model patient simulators are now used to train counsellors (Wang et al., 2024b; Louie et al., 2026), to evaluate LLM therapists and psychiatric interview agents (Wang et al., 2024a; Lee et al., 2025; Yang et al., 2026), and are built from clinical theory or from real patient histories (Sabour et al., 2026; Li et al., 2026). A randomised study of more than ninety novice counsellors found that practising with simulated clients changes measured skill (Louie et al., 2026), so the fidelity of simulated patients affects the people trained with them.

Patients with the same severity report different symptoms (Fried and Nesse, 2015a): one sleeps badly and has lost interest in things, another feels worthless and cannot concentrate. Specific symptoms differ in risk factors, impairment and course (Fried and Nesse, 2015b), and they occur in groups, such as somatic and non-somatic symptoms of depression (Elhai et al., 2012). Training with simulated clients is meant to expose trainees to this range (Wang et al., 2024b), and an agent that assesses patients has to recognise which symptoms a patient has as well as how severe they are. A simulator lacking this variation offers trainees and tested tools fewer presentations than clinicians meet, although each transcript can be plausible.

Simulators are evaluated mainly by clinicians rating transcripts for plausibility (Wang et al., 2024b; Hoang et al., 2026; Yang et al., 2026) and by comparing simulated total scores with population data (Keough, 2026), which leaves two comparisons open. First, language models used as survey respondents vary less than the people they stand in for (Bisbee et al., 2024; Petrov et al., 2024), but no evaluation asks whether simulated patients with the same total score differ from one another, item by item, as much as real ones. Second, clinical psychology studies imitation by instructing healthy people to feign a disorder and comparing their answers with those of patients (Rogers and Bender, 2020); simulators have not been compared with such people, so it is unknown whether models imitate patients as people do.

We first give a framework for simulated respondents. An information budget bounds how much simulated patients can differ, and an average-

patient hypothesis holds that, given only a persona, a model answers each item at the level typical of the named condition and severity. Together they predict how simulated patients differ from real respondents with the same total, and the budget limits what changes to the persona can add. We examine these implications for 17 models on public item-level data for eight questionnaires on seven kinds of symptoms from eight human samples, including a clinical sample and a study with an instructed-feigning group, and make both comparisons at matched totals, which separates how a total is composed from how high it is. We also use the personas of two published simulators and examine clinical interviews and an assessment agent.

Our contributions are as follows:

- **Simulated patients answer like an average patient.** Across 17 models, simulated patients vary 1.5 to 11 times less than real respondents with the same total and on most questionnaires endorse nearly every symptom at a middling level, as an information budget and an average-patient hypothesis predict.
- **Common remedies reduce the pattern without removing it, and it limits what simulated patients can test.** Prompt changes, published personas, larger models and reasoning all leave simulated patients less varied than real respondents. In interviews with severity personas, an assessment agent's recognition of absent symptoms goes untested and its measured accuracy against the stated severity halves.
- **Real symptom profiles mitigate the pattern.** Having the model word a real profile restores half to two thirds of the variation in interviews and makes absent symptoms testable.

These results bear on training clinicians with simulated patients, on benchmarks that evaluate clinical agents with them, and on research that uses language models to simulate survey respondents or study participants: all of these need simulated individuals to differ as real ones do.

2 Related Work

Simulated patients and their evaluation. Besides clinician ratings of plausibility (Wang et al., 2024b; Hoang et al., 2026; Yang et al., 2026), a population audit compares simulated total scores and screening rates with epidemiological data across demographic groups (Keough, 2026), simulated eating-disorder patients reach ceiling scores (Haase et al., 2026), and a workshop study finds that the PHQ-9 and GAD-7 items of four open-weight models do not function as in survey respondents (Shen and Han, 2026). Unlike these evaluations, we test whether simulated respondents with the same total differ as much as real ones, compare simulators with people instructed to feign (Rogers and Bender, 2020) and show the effect on evaluating an assessment agent.

Language models as survey respondents. Language models used as synthetic survey respondents reproduce some aggregate opinions of subpopulations (Argyle et al., 2023), but their opinion distributions differ from those of the groups they represent (Santurkar et al., 2023), vary less than real responses and change with prompt wording (Bisbee et al., 2024), follow option order and labels (Dominguez-Olmedo et al., 2024), and match individuals worse than aggregates (Petrov et al., 2024). We find the same homogeneity on clinical questionnaires, which allow three further steps: holding the total fixed separates how a response is composed from how high it is, people instructed to feign show how humans imitate a patient, and per-item answer distributions show the source of the homogeneity. Appendix A gives a fuller discussion.

3 A Framework for Simulated Respondents

3.1 Quantities at a fixed total

A response to a K-item questionnaire is $X = (X_1, \ldots, X_K) \in \{0, \ldots, m\}^K$ with total $T = \sum_k X_k$; the PHQ-9 has $K = 9$ and $m = 3$. Real respondents follow a distribution p. A simulator receives a persona $\Pi \sim P_\Pi$, a text that names a severity s and may add a biography or a symptom profile, and answers $\hat{X} \sim q(\cdot \mid \Pi)$; item order and sampling enter through q. For a group r, real or simulated, let X and X' be independent draws from r given $T = t$. At this total we define breadth, the share of "not at all" answers, the share of responses endorsing every item, and diversity:

$$B_r(t) = \textstyle\sum_k \Pr(X_k > 0),$$
$$Z_r(t) = \tfrac{1}{K} \textstyle\sum_k \Pr(X_k = 0),$$
$$A_r(t) = \Pr(\min_k X_k > 0),$$
$$D_r(t) = \mathbb{E}\|X - X'\|_1.$$

The first three describe composition, which symptoms carry the total, and D_r the amount of variation.

Structure is the correlation matrix ρ_r of the residuals $X - \mathbb{E}_r[X \mid T]$. Against the real reference ρ_p we measure its similarity and the contrast between the somatic and the non-somatic items (Elhai et al., 2012),

$$S_r = \mathrm{corr}_{j<k}\big(\rho_{r,jk}, \rho_{p,jk}\big),$$
$$C_r = \bar{\rho}_r^{\text{within}} - \bar{\rho}_r^{\text{between}},$$

where the means run over pairs of items within the same group and across the two groups. Separability is the AUC of a logistic regression $\sigma(\beta_0 + \sum_k \beta_k x_k)$ that tells simulated from total-matched real responses.

Proposition 1. *With $r_{t,k}$ the distribution of X_k given $T = t$,*

$$D_r(t) = \textstyle\sum_k \sum_{a,b} |a - b|\, r_{t,k}(a)\, r_{t,k}(b).$$

Proposition 2. *If $q_{t,k} = p_{t,k}$ for every t and k, the best-fitting logistic regression on item-wise features of total-matched samples is constant, with AUC $1/2$.*

Diversity thus depends only on the item distributions at each total, an additive classifier cannot detect a difference confined to how items covary, and S_r and C_r measure that covariation.

3.2 Information budget

Proposition 3 (Information budget)**.** *For any persona distribution and simulator,*

$$H(\hat{X}) \le I(\Pi; \hat{X}) + \textstyle\sum_k H(\hat{X}_k \mid \Pi)$$
$$\le H(\Pi) + \textstyle\sum_k H(\hat{X}_k \mid \Pi).$$

Answers to one persona π have perplexity at most $2^{h(\pi)}$, and two of them are identical with probability at least $2^{-h(\pi)}$, where $h(\pi) = \sum_k H(\hat{X}_k \mid \Pi = \pi)$.

Simulated patients can therefore differ only through information that the persona carries into the answers, $I(\Pi; \hat{X})$, or through the randomness of single answers. A persona that names one of four severities has $H(\Pi) \le 2$ bits, and 400 client descriptions have up to $\log_2 400 \approx 8.6$ bits, enough for each client to answer differently.

3.3 The average-patient hypothesis

We hypothesise that a model answers each item at the level typical of that item for the named condition and severity: for a persona π that names severity s, there are levels $\ell_k(s) \in \{0, \dots, m\}$ and an $\varepsilon \le \frac{1}{2}$ with

$$q(\hat{X}_k = \ell_k(s) \mid \pi) \ge 1 - \varepsilon \quad \text{for every item } k.$$

When every item is a core symptom of the named condition, as on the PHQ-9 for depression, one level $\ell(s) \ge 1$ serves all items. The persona is the model's only information, and instruction-tuned models give concentrated answers to survey questions (Santurkar et al., 2023).

Proposition 4 (Average patient)**.** *Under the hypothesis, answers to π satisfy*

$$H(\hat{X}_k \mid \pi) \le h_2(\varepsilon) + \varepsilon \log_2 m,$$
$$\mathbb{E}\|\hat{X} - \hat{X}'\|_1 \le 2mK\varepsilon,$$
$$\Pr\big(\hat{T} = \textstyle\sum_k \ell_k(s)\big) \ge 1 - K\varepsilon,$$

and, if every $\ell_k(s) \ge 1$, $A \ge 1 - K\varepsilon$ and $Z \le \varepsilon$, where h_2 is the binary entropy and $\hat{X}, \hat{X}'$ are independent answers to π.

Simulated patients should therefore vary little, endorse every item whose typical level is above zero, and give totals near $\sum_k \ell_k(s)$: on the PHQ-9, 9 for mild, 18 for moderate and 27 for severe depression, at the top of the bands 5–9 and 20–27 and above the band 10–14. The hypothesis and Proposition 4 concern the answers to one persona; we check the hypothesis on items asked one at a time, and the comparisons at matched totals, where the measures condition on the total, test its predictions empirically. The hypothesis leaves ρ_q unconstrained, and independent noise raises D_q without restoring ρ_p. By Proposition 3, the budget grows only with $I(\Pi; \hat{X})$ or the entropy of single answers, and within it item dependence also changes the joint entropy. Appendix E gives the proofs.

4 Data and Method

4.1 Instruments and human references

Main instruments. The PHQ-9 (Kroenke et al., 2001) rates nine depressive symptoms over two weeks from 0, not at all, to 3, nearly every day; item 9 asks about thoughts of being better off dead or of self-harm. NHANES[1] records it for N=41,714 adults in eight cycles from 2005 to 2023. The GAD-7 (Spitzer et al., 2006) rates seven anxiety symptoms on the same scale; the 2019 NHIS[2] Sample

[1] https://wwwn.cdc.gov/nchs/nhanes/
[2] https://www.cdc.gov/nchs/nhis/

Adult file records it for N=31,132 adults, with the PHQ-8 (Kroenke et al., 2009) to check NHANES.

Feigning study. A public OSF study (Boskovic and Giromini, 2026) gives the PCL-5 (Blevins et al., 2015), 20 post-traumatic symptoms rated 0–4, for university students in three groups: 69 genuinely injured, the genuine patients below, 46 uninjured answering honestly, and 39 uninjured told that an injury would earn a tuition deduction they are not entitled to and that "no one would ever find out".

Extension. Five more datasets from the Item Response Warehouse (Domingue et al., 2025) add seven references: the K6 distress scale (Kessler et al., 2002) in NHIS 2016–2018; the OCI-R (Foa et al., 2002), the PHQ-15 somatic scale (Kroenke et al., 2002) and the ITQ post-traumatic items (Cloitre et al., 2018) from a UK quota sample (C19PRC; McBride et al., 2021); the EDE-Q (Fairburn and Beglin, 1994) in the Mexican population (Contreras-Valdez et al., 2022); the PHQ-9 at intake to a depression trial in rural Chinese adults aged 60 or more (COACH, clinical; Chen et al., 2018); and the PCL-5 in Greek women after caesarean section (obstetric; Orovou et al., 2021). Table 1 gives sample sizes, Appendix B sources and preprocessing.

4.2 Models and administration

We query 17 models: GPT-4o-mini, GPT-4o, GPT-4.1-nano, GPT-4.1-mini, GPT-4.1, GPT-5-mini, GPT-5.2, GPT-5.4-mini, Claude Haiku 4.5, Grok-4.1-fast, Llama-3.3-70B, Kimi-K2.5, Mistral-Large-3, Phi-4, and Gemini 2.5 Flash-Lite, Flash and Pro, with reasoning off or at its minimum; Appendix C gives the settings and Appendix D the runs with reasoning on. Each model completes each questionnaire in nine cells, following the simulation design used with human feigners (Rogers and Bender, 2020): an honest control plus {naive, coached} × {mild, moderate, moderately severe, severe}, with 50 trials per cell and the item order randomised per trial. Each trial gives the model a one-line persona naming the severity, as in population audits (Shen and Han, 2026), and the coached arm adds that questionnaires contain checks for exaggeration; Appendix C gives the prompts. We also use the personas of two established simulators without their dialogue frameworks: the 12 clinician-validated depression personas of TalkDep (Wang et al., 2025) and the 400 client intake forms of CACTUS (Lee et al., 2024), alone and with each client's case formulation. In the feigning study, models take the three roles with matched injuries.

4.3 Estimation

Each model is compared with real respondents at exactly its totals: on the PHQ-9 and GAD-7 at totals 15–24 and 12–19, where both groups have ample responses, and on the nine references at every total that real respondents reach, in narrow bins on long scales. Residual correlations use totals 5–24, with all NHANES respondents as ρ_p, and separability is the 5-fold cross-validated AUC against total-matched real responses. Item entropies come from the 50 whole forms of each severity cell and, for ten models, from log-probabilities when an item is asked alone. A one-factor graded response model (GRM) fit to NHANES 2005–2012, whose items are independent given one trait, gives profiles without the population's residual covariation. Appendix D gives intervals and permutation tests (Ojala and Garriga, 2010).

5 Response Profiles at Matched Total Score

5.1 PHQ-9 and GAD-7

Composition. As the hypothesis predicts, Figure 1a shows that at PHQ-9 totals 15–24 every model endorses more symptoms at lower intensity than real respondents with the same totals. In Figure 1b, models answer "not at all" in 0–4.5% of answers, against 10–15% for real respondents. Figure 1c shows that models endorse all of items 1–8 in 99.1–100% of responses and all seven GAD-7 items in 99–100%, against 47–62% of real respondents with the same totals. Figure 6d,e and Table 6 show the same in every band with symptoms.

Amount of variation. In Figure 1d, two responses from one model differ 3.0–8.6 times less than two real respondents with the same total. Item scores alone separate simulated from total-matched real responses with an AUC of 0.68–0.86 on the PHQ-9 and 0.63–0.77 on the GAD-7, against 0.56 for two real samples.

Structure. Residual correlations at a fixed total depend on the item distributions as well as on covariation, so we also draw responses with each model's item distributions at each total and no further covariation. Real samples drawn at each model's totals agree with NHANES at $S = 0.53$–0.77, the models at 0.13–0.61 and the responses without covariation at -0.02 to 0.35. The models thus covary beyond their item distributions, 11 of 17 above the 97.5th percentile of that reference, but less like the population than real respondents do,

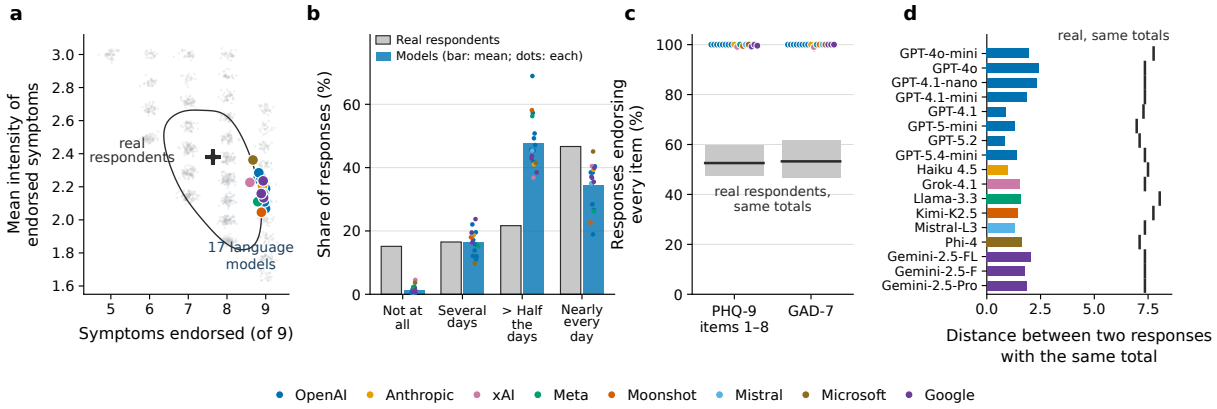


Figure 1: **Response profiles on the PHQ-9 and GAD-7.** Panels a, b and d use PHQ-9 responses with total 15–24. **a**, Symptoms endorsed against their mean intensity (grey dots, real respondents; contour, their densest half; coloured, model means). **b**, Use of the response options. **c**, Responses endorsing every item, PHQ-9 items 1–8 (total 15–24) and GAD-7 (12–19), with 95% intervals; grey, real respondents at each model's totals (range and median over models). **d**, Diversity; ticks, real respondents at each model's totals.

						All items (%)	
Instr.	Sample	N	Cov.	Div. ratio	AUC	Real	Models
PHQ-9	NHANES	41,714	100	3.5–11.2	0.66–0.83	25	40–86
PHQ-9	COACH†	2,359	84	3.2–8.2	0.72–0.91	22	31–89
GAD-7	NHIS	31,132	100	3.9–10.7	0.65–0.76	45	71–100
K6	NHIS	82,339	100	2.5–10.0	0.60–0.80	63	83–100
PHQ-15	C19PRC	2,025	74	2.1–5.4	0.80–0.97	41	2–58
OCI-R	C19PRC	1,406	72	2.0–4.2	0.77–0.99	51	5–98
ITQ	C19PRC	2,025	100	1.5–4.3	0.63–0.79	74	81–100
PCL-5	Postpartum†	469	44	—	0.85–0.97	14	15–77
EDE-Q	Mexico	2,090	78	2.7–5.5	0.88–0.96	18	71–100

Table 1: Profile comparison on nine human references over all totals, each simulated response matched to real respondents with the same total (ranges over models). Cov.: % of simulated responses whose total real respondents reach. Div. ratio: real over model diversity, where five or more totals carry it (none on the postpartum PCL-5; Table 11). AUC: separability from matched real respondents (chance 0.46–0.52). All items: % endorsing every item. Real and Cov.: medians over models. †Clinical (COACH) and obstetric (postpartum) samples.

16 of 17 below the real samples; Tables 36 and 37 give the values. Adding independent noise until the models are as varied as real respondents lowers S to 0.01–0.12. The GRM gives $S = 0.40$ with 20,000 profiles and no somatic–non-somatic contrast, although Table 30 shows that it is as varied as held-out 2013–2023 respondents and as hard to separate from them, as Propositions 1 and 2 imply.

Stated severity. Told they are mildly, moderately, moderately severely or severely depressed, the models give median totals of 12, 17, 21 and 25, near the $9\ell(s)$ of Proposition 4 for moderate and severe depression and above it for mild. Table 38 shows that each model's total lies above the band of a mild to moderately severe label in 61–100% of naive-arm responses. A label may also be read as a diagnosis, for which a total of 10 or more is the usual threshold (Kroenke et al., 2001), but told only of mild daily-life impact, both models tested exceed the mild band in 94–98% of responses.

5.2 Nine human references

Table 1 and Figure 14 apply one total-matched protocol to all nine references. Real respondents with the same total are 1.5–11 times more varied than every model wherever at least five totals carry the estimate, and every model is separable from them with an AUC of at least 0.60, against 0.49–0.56 between real samples. Models endorse every item more often on seven references, including the clinical sample, whose patients endorse all nine items less often than NHANES respondents with the same totals, 2.5% against 5.0%. On the PHQ-15 and the OCI-R, Figure 15 shows that models endorse fainting spells, pain during intercourse and hoarding less often than real respondents, 21% against 76% for fainting, and most other items more often, consistent with levels $\ell_k(s)$ of zero for atypical symptoms.

5.3 Comparison with instructed feigners

As Figure 2a and Table 13 show, people told to feign an injury for money answer "not at all" 0.90 times as often as genuine patients with the same totals (95% interval 0.75–1.03), and uninjured students answering honestly 0.99 times as often (0.91–1.09). The 17 models playing a genuinely injured student answer "not at all" at most 0.37 times as often. On a log-odds scale the feigners' odds ratio

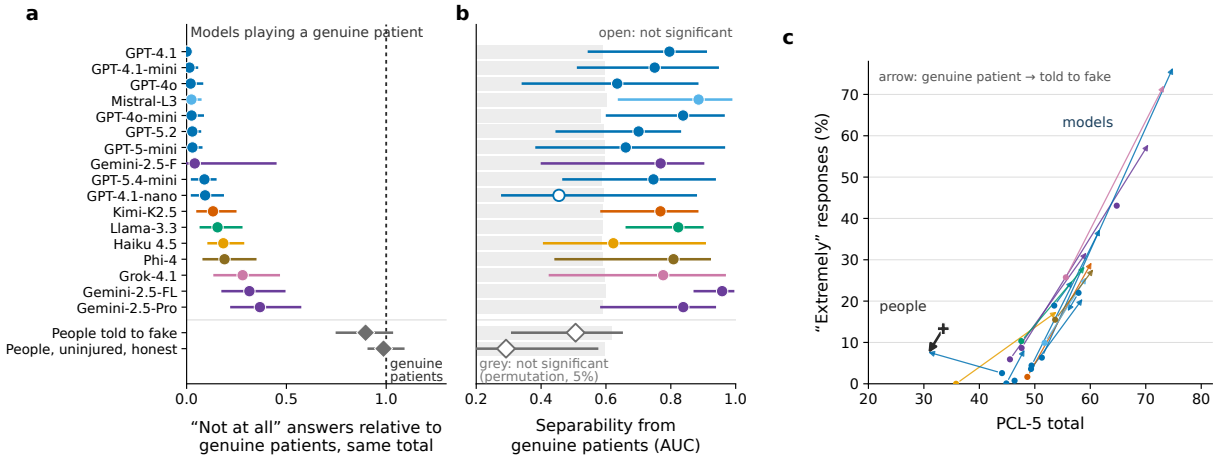


Figure 2: **Comparison with people instructed to feign** (PCL-5). **a**, "Not at all" answers relative to genuinely injured students with the same totals (95% bootstrap intervals). **b**, Separability from them, with the 2.5th–97.5th percentiles over 80% subsamples; grey, below each group's one-sided 5% critical value of a label-permutation test; open, not significant (Holm-adjusted for models). **c**, Change in total and in "extremely" answers from patient to feigning role.

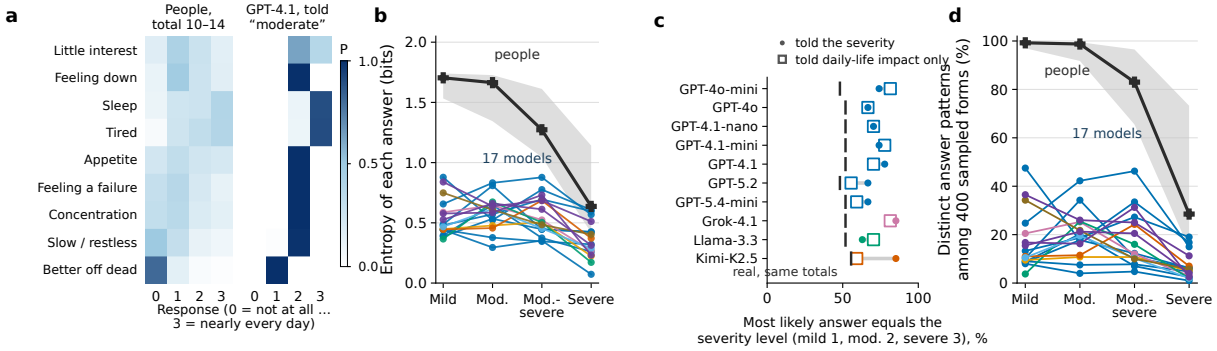


Figure 3: **Per-item answer distributions. a**, Answer probabilities per PHQ-9 item: real respondents with total 10–14 and GPT-4.1 told "moderate" (log-probabilities). **b**, Entropy of each item's answers across the whole-form responses of a severity cell. **c**, Items whose most likely answer, from log-probabilities, equals the severity level, told the severity (dots) or only how daily life is affected (squares); ticks, real respondents with the same totals as the model's whole forms; moderately severe is left out. **d**, Distinct profiles among 400 forms sampled item by item from the answers in b. People in b, d: real respondents at each model's totals (grey, range over models; line, median).

is 0.71, so they too answer "not at all" somewhat less often than genuine patients, and every model's is 0.26 or less, below the feigners' after Holm adjustment. Honest and feigning people therefore compose a total closer to that of genuine patients than any model. Figure 2c shows that, told to feign, people do not raise their total, whereas 15 of 17 models raise both their total and their share of "extremely" answers. In Figure 2b, a classifier separates 16 of 17 models from genuine patients and does not detect the feigners, whose AUC is 0.51, although with 14–24 genuine patients per model this test is weak.

6 Per-Item Answers, Prompts and Scale

6.1 Per-item answer distributions

Figure 3a,b and Table 15 show that, as the hypothesis requires, single answers are highly concentrated: their entropy averages 0.29–0.73 bits for the models, out of at most 2, against 1.15–1.51 bits for real respondents with the same totals. At the average item entropy, the perplexity bound of Proposition 3 is 6–97 for the models and about 1,300–12,500 for real respondents with the same totals. In Figure 3d, forms drawn item by item from these distributions are at most 32% distinct on average, against 69–92% for the same draws from real respondents. In Figure 3c, the most likely answer is $\ell(s)$ for 63–85% of items, also without the severity word,

Condition	Diversity (%)	N.a.a.	AUC	Calib.
Real respondents, held out	100	9.9	—	—
One-line persona	16 (9–29)	0.0	0.81 (0.64–0.88)	39
+ "people differ"	29 (26–44)	0.1	0.84 (0.76–0.87)	48
+ biography	23 (17–29)	0.0	0.80 (0.71–0.86)	47
Target total, no label	35 (27–51)	0.2	0.69 (0.59–0.82)	68
+ 5 real answer sheets	48 (31–57)	0.5	0.68 (0.57–0.69)	83
TalkDep personas	41 (32–50)	0.0	0.86 (0.75–0.89)	—
CACTUS personas	39 (33–42)	1.1	0.88 (0.84–0.92)	—
+ case formulation	37 (32–40)	1.2	0.91 (0.86–0.93)	—

Table 2: Prompt changes and published personas, PHQ-9, seven models: median (range). Diversity: % of held-out real respondents with the same totals (totals 5–27). N.a.a.: "not at all" answers, items 1–8 at totals 15–24 (%). AUC: separability from held-out real respondents (totals 15–24). Calib.: % of totals in the instructed band; the personas name no band. TalkDep: pairs of different personas, a persona in one fold.

against 48–56% for real respondents.

6.2 Prompt changes, personas and scale

Table 2 shows that for the seven models in Table 3, stating that people with the same severity differ, adding a biography or replacing the severity word with a target total changes separability from held-out real respondents by -0.08 to $+0.03$ on average. The information budget allows a biography or a target total to help, since they add information to the persona, so their small effect shows that little of this information reaches the composition of the answers. Five real answer sheets from the target band, which add information about symptoms, lower separability most, by 0.14, and raise diversity to 31–57% of the real value at the same totals. The 400 CACTUS clients, each answering once, carry up to 8.6 bits, but the 17 models give 43–68% distinct answer patterns to them on the PHQ-9, against 97–98% for real respondents with the same totals. For TalkDep, whose 12 personas are answered repeatedly, we compare only answers of different personas, and Tables 21 and 23 show that with the published personas all 17 models stay 1.6–4.8 times less varied than real respondents on both questionnaires. Figure 9d,e and Table 24 show that larger tiers in a family do not reliably raise diversity and that none of 52 configurations exceeds 57% of real diversity.

7 Interviews and Downstream Use

7.1 Simulated clinical interviews

We interview simulated patients with a fixed script covering the nine PHQ-9 domains, so that every item can be rated. GPT-4.1 and Claude Haiku 4.5 each code the PHQ-9 from the transcript, neither for its own patients, and seven patient models give 80 interviews with a severity persona and 84 with the 12 TalkDep personas. Interviews with a severity persona are uniform: Figure 4a and Table 25 show that 97.5–100% are rated as endorsing all eight non-suicidality symptoms, against 49–85% of real respondents with the same totals, and Figure 4b that they differ little from one another. Real profiles passed through the same interview are also compressed, but Table 27 shows that severity-persona interviews remain 2.9–4.6 times less varied than these, and Table 28 that TalkDep interviews of different personas remain 1.6–2.2 times less varied; with personas labelled moderate or above, 81–100% are rated as having all eight symptoms, against 45–80% of real respondents at those totals.

7.2 Profiles from real data or an item response model

Given a real NHANES profile as private background, five patient models give interviews whose rated answers follow the given ones with an item correlation of 0.70–0.83, as Figure 4c shows. In Figure 4b, these interviews differ from one another more than those with a severity persona, with a diversity of 3.30–4.91 against 0.36–1.01, which recovers half to two thirds of the diversity of the given profiles. Table 25 shows the same for profiles sampled from the GRM, which can be used where real profiles cannot. Five graduate students without clinical training, shown no condition labels or profiles, rated 64 transcripts. Table 29 shows that they agree with one another, with a weighted κ of 0.79–0.87, and rate all eight symptoms as present in each of the 14 severity-persona interviews. They rate symptoms given as "several days" higher in 75% of cases, against 91% for the model rater, so part of this upward shift lies in the patients' wording.

7.3 Evaluating an assessment agent

We treat the interviewer and the model rater as an assessment agent and score it against what each simulated patient was given, as benchmarks do (Lee et al., 2025; Wang et al., 2025). Uniform patients leave symptom recognition untested. Patients given a real profile lack 18.8% of symptoms 1–8, and the agent matches the profile on 96–98% of symptoms, against 81% for an agent that reports every symptom present. Figure 4a shows that patients told a severity are rated as lacking almost

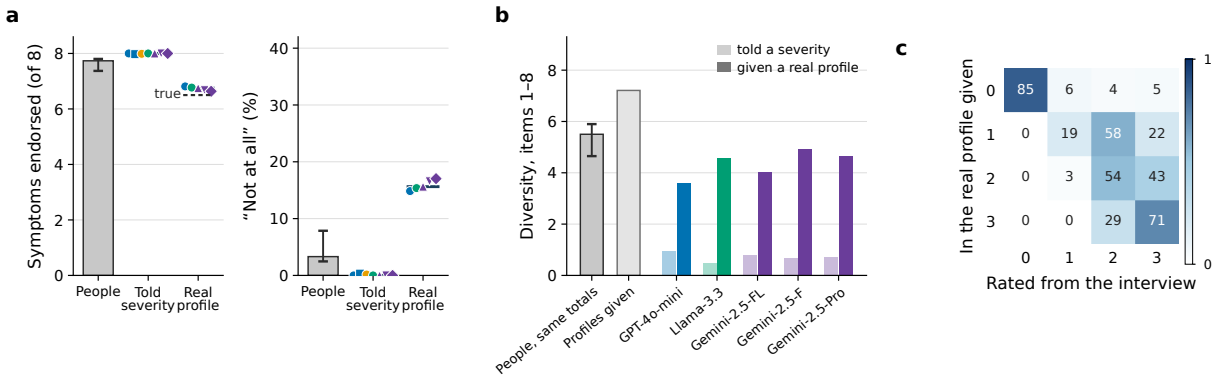


Figure 4: **Simulated clinical interviews. a**, Symptoms endorsed and "not at all" answers, PHQ-9 items 1–8 rated from transcripts (dots, patient models coloured as in b; line, median; people, real respondents with the same totals; dashed, the profiles given). **b**, Diversity. **c**, Answer in the real profile given against the answer rated from the interview, row percentages.

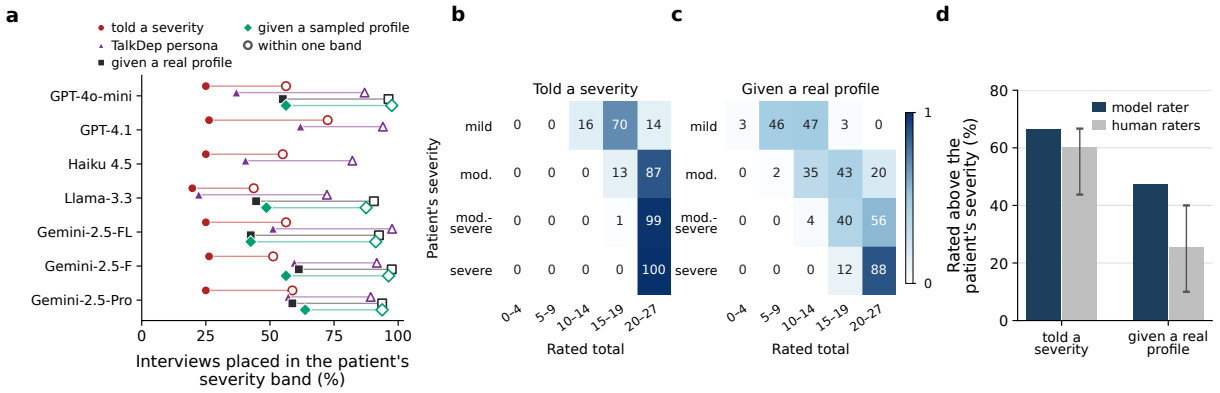


Figure 5: **Severity rated by an assessment agent. a**, Interviews whose rated PHQ-9 total falls in the patient's severity band (filled) or within one band (open), per patient model and condition; for TalkDep personas, the band named like the persona's severity label. **b**, **c**, Rated total against the patient's severity, row percentages. **d**, Interviews rated above the patient's severity by the model rater and by five human raters on the 64 transcripts they rated, severities weighted equally (range over raters).

none of their symptoms, 0–0.3%, so on them the two agents cannot be told apart.

Figure 5a shows that the agent places 42–61% of patients given a real profile in their severity band and 20–26% of patients told a severity. In Figure 5b,c, patients told they are mildly to moderately severely depressed are rated above that band in 99–100% of interviews, so only those told they are severe are placed correctly, and scoring the agent against a stated severity halves its measured accuracy, whether the patient overstates the label or reads it as a diagnosis. TalkDep personas, which state the severity of each key symptom, are placed in their labelled band in 22–62% of interviews, about as often as patients given a real profile, and the agent ranks them by BDI-II score with a Spearman correlation of 0.87–0.95. On 64 transcripts with severities weighted equally, Table 35 shows that human raters place 35% of patients told a severity and 56% of those given a real profile in their band, against 33% and 48% for the model rater, and Figure 5d that they rate 60% and 26% above it. In Appendix D, a screening model trained on simulated patients is miscalibrated and less accurate than one trained on total-matched real respondents.

8 Conclusion

Across 17 models and nine references, simulated patients behave as our framework predicts: they spread a total over more symptoms than real respondents with the same total, vary less and exceed the PHQ-9 band of their stated severity, and prompt changes, published personas, larger models and reasoning do not remove this. We recommend evaluating simulators item by item at matched totals, including covariation, scoring assessment agents against a given profile, and grounding simulated patients in real symptom profiles.

Limitations

The human references are public survey, clinical and student samples that differ from the models' personas in population, language and mode of administration, and the comparison with feigners rests on one public study of post-injury symptoms. The interviews follow a fixed script and are rated by models, with human raters checking a subset, and the assessment agent is our own interviewer and rater; published simulators run with their dialogue frameworks, and other agents, may behave differently. The severity personas use the names of the PHQ-9 bands, which a model may also read as diagnoses, and the analysis of covariation uses the PHQ-9. We study models as simulators use them, through prompts and personas; how training, such as instruction tuning, gives rise to the concentration of single answers is left for future work.

Ethics Statement

We use de-identified public data released for secondary analysis, and no participants were recruited for the questionnaire comparisons. Five graduate students volunteered, without payment, to rate model-generated interview transcripts; they were told beforehand that some transcripts contain role-played thoughts of death or self-harm and that they could stop at any time. Prompts ask models to role-play psychiatric symptoms, including suicidal ideation, inside an explicit clinical-training frame, and no output reached a user. The results concern systems used to train clinicians, and the code is released so that simulators can be evaluated. Sampling profiles from real data could also make feigned questionnaires harder to detect; we report no method aimed at evading validity checks.

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A Extended Related Work

LLM patient simulators for mental health. Wang et al. (2024b) build PATIENT-ψ, simulated clients constructed from CBT cognitive models and validated by clinicians, and show that practising against them changes what trainees notice. Later systems vary the source of the persona: Sabour et al. (2026) derive profiles from clinical theory, Lee et al. (2025) generate them for structured psychiatric assessment, and Li et al. (2026) ground them in longitudinal social-media histories aligned to diagnostic interviews. Simulated clients are also used to evaluate LLM therapists (Wang et al., 2024a). Louie et al. (2026) run a randomised study with more than ninety novice counsellors and find simulated practice improves measured skill, so the fidelity of the simulation matters for training.

Evaluation of these systems relies mainly on expert judgement. Hoang et al. (2026) build a clinically grounded benchmark for depression patient simulators with high expert agreement, and Yang et al. (2026) construct a virtual psychiatric environment with clinician-validated evaluators. These benchmarks measure whether a transcript looks plausible to a rater and do not compare simulated responses with item-level data from real respondents.

Language models as survey respondents. Models have been given standard personality inventories (Serapio-García et al., 2023) and used as synthetic survey respondents whose aggregate opinions track real subpopulations (Argyle et al., 2023). Individual responses match less well: opinion distributions differ from those of the demographic groups a model is asked to represent (Santurkar et al., 2023), synthetic responses vary less than real ones and change with prompt wording (Bisbee et al., 2024), answers follow option order and labels and approach uniform randomness once these are adjusted for (Dominguez-Olmedo et al., 2024), and reproduction is better at the aggregate than the individual level, with respondents too homogeneous and too internally consistent (Petrov et al., 2024). Our diversity result extends these findings to clinical questionnaires, and the person-fit result (Table 7) shows the same excess consistency. Clinical questionnaires are scored, so holding the total fixed separates how a response is composed from how high it is, which population comparisons cannot do. People instructed to feign show how humans imitate a patient, people answering honestly show that the comparison separates models from people and does not grade imitations, and per-item log-probabilities show that each answer is concentrated at the severity the prompt implies.

Closest to this work, Keough (2026) audit the

epidemiological fidelity of LLM mental-health simulation across 120 demographic cohorts. Against NHANES they compare PHQ-8 means and screening rates, which the simulations overestimate for most groups, and the marginal shares of answer categories; within the simulated data they find that elevated cases mostly satisfy the DSM-5 gateway rule, that repeated draws of the same prompt often land in different severity categories, and that item correlation matrices differ between cohorts. Their comparisons with real data concern totals, screening rates and marginal answer shares without holding severity fixed, and an earlier version's variance-compression result was withdrawn. Shen and Han (2026) give four open-weight models severity personas on the PHQ-9 and GAD-7 and compare them with NHANES and NHIS respondents by confirmatory factor analysis and differential item functioning: the models are highly internally consistent, fail scalar invariance and show differential functioning on most PHQ-9 items. Differential item functioning compares item means at the same total and leaves open how much respondents with the same total differ from one another. We compare each simulated response item by item with real respondents who have the same total, measure how much responses with the same total differ, add a human feigning reference, and trace the result to per-item answer distributions; the deviation we find is not visible in total-score distributions. Haase et al. (2026) simulate eating-disorder patients on the 28-item EDE-Q, find scores that saturate at ceiling and fail to preserve severity rankings, and propose testing whether this extends to behaviourally graded instruments such as the PHQ-9. We run that test on the PHQ-9 and eight other references, including the EDE-Q against real respondents.

Response-pattern measurement. Psychometrics has long studied aberrant response patterns. Drasgow et al. (1985) introduce the standardised log-likelihood l_z for polytomous models; Snijders (2001) shows l_z is biased when the person parameter is estimated, so we compare l_z with its distribution among real respondents. Emons (2008) develops nonparametric person-fit for polytomous items, including the Guttman-error statistic we normalise by its maximum at a given total. The graded response model (Samejima, 1969) supplies the item parameters (Tables 8 and 9).

Feigning. Clinical psychology detects feigned illness with a small set of strategies: endorsement of rare symptoms, of improbable symptom combinations, and of symptoms at implausible severity (Rogers and Bender, 2020). Such instruments are validated with simulation designs, in which healthy participants are instructed to feign and compared with genuine patients; coaching participants about validity checks reduces but does not remove their detectability. The standard instruments are commercial and restricted, so we implement the published strategies against public base rates, and use the one public item-level simulation study we found (Boskovic and Giromini, 2026) for the direct comparison.

B Human Reference Samples

NHANES (PHQ-9). The depression screener module records the nine items on 0–3. We pool cycles 2005–06 through 2017–18 and 2021–23, restricted to adults with no missing, refused or don't-know answers (N=41,714); the 2017–March 2020 file is excluded because it overlaps 2017–18. Mean PHQ-9 is 3.32 (s.d. 4.34) and 9.33% score at least 10, matching published estimates. The pandas XPORT reader decodes SAS numeric zero as a denormal ($\approx 5.4 \times 10^{-79}$); untreated, every zero fails the validity filter and the sample shrinks to 444 respondents with mean 18.5. We normalise these values at read time and abort if the complete-case filter would remove more than 35% of respondents.

NHIS 2019 (GAD-7, PHQ-8). Items are coded 1–4 and recoded to 0–3 (N=31,132 and 31,066). NHIS is independent of NHANES and uses a different administration mode. Item endorsement rates of the PHQ-8 agree with the first eight PHQ-9 items in NHANES at r=0.99, and breadth and unevenness agree in every score band (Figure 13a,b).

OSF feigning study (PCL-5). University students (mean age about 21) completed the PCL-5 in one of three groups (genuinely injured, n=69; uninjured and honest, 46; uninjured and instructed to feign an injury to obtain a tuition deduction, 39) (Boskovic and Giromini, 2026). Among the injured, 68% reported a psychological injury; feigners chose their own injury type (57% psychological). The file ships a PCL-5 rare-items index. We recovered its scoring and reproduce it exactly for all 154 respondents: one point each for items 3, 9, 10 and 15 at 4 and items 8 and 16 at $\geq$3; flag at $\geq$3.

Extension samples. All were located through the Item Response Warehouse (Domingue et al., 2025)

and downloaded from the original sources without registration.

- K6 (Kessler et al., 2002): NHIS Sample Adult 2016–2018, items coded 1 (all of the time) to 5 (none), recoded 0–4; N=82,339.
- C19PRC-UK (McBride et al., 2021): UK adults recruited in 2020 by quota on age, sex and household income. OCI-R (Foa et al., 2002) from wave 2 (N=1,406); PHQ-15 (Kroenke et al., 2002) without the menstrual item (14 items, 0–2) and the six ITQ post-traumatic symptom items (Cloitre et al., 2018) from wave 1 (N=2,025). The survey anchored the ITQ to respondents' experience of the COVID-19 pandemic; our prompts use the same anchor.
- EDE-Q 6.0 (Fairburn and Beglin, 1994): Mexican general population, two studies pooled (N=2,090; Contreras-Valdez et al., 2022). We use the 22 attitudinal items scored 0–6 (items 1–12 and 19–28); items 13–18 are behaviour counts.
- COACH (Chen et al., 2018): baseline PHQ-9, administered by primary-care physicians, for adults aged 60 or more in rural Chinese villages with depression and hypertension enrolled in a cluster randomised trial (inclusion PHQ-9 $\geq$10; N=2,359, mean 12.1).
- Postpartum PCL-5 (Orovou et al., 2021): Greek women after caesarean section (N=469, median total 0).

C Models, Prompts and Administration

Models. OpenAI models were queried through the OpenAI API or Azure AI Foundry, Gemini models through the Google AI Studio API, and all others through Azure AI Foundry. Deployments: `gpt-4o-mini`, `gpt-4o`, `gpt-4.1-nano`, `gpt-4.1-mini`, `gpt-4.1`, `gpt-5-mini`, `gpt-5.2`, `gpt-5.4-mini`, `claude-haiku-4-5`, `grok-4-1-fast-non-reasoning`, `Llama-3.3-70B-Instruct`, `Kimi-K2.5`, `Mistral-Large-3`, `Phi-4`, `gemini-2.5-flash-lite`, `gemini-2.5-flash`, `gemini-2.5-pro`. Temperature was 1.0 where the API accepts it; the GPT-5 series and the Anthropic SDK do not, and run at their defaults. Answers use a JSON schema with one required integer per item; Phi-4 rejects JSON schemas and was run in JSON mode with the same validation. Reasoning is off for every model except Gemini 2.5 Pro, whose API rejects a zero thinking budget and which runs at its minimum budget of 128 thinking tokens (set through the Google extension of the OpenAI-compatible endpoint): GPT-5-mini runs at reasoning effort minimal and Gemini 2.5 Flash and Kimi-K2.5 at none, and none of their calls produces reasoning tokens; GPT-5.2, GPT-5.4-mini and Gemini 2.5 Flash-Lite do not reason at their defaults, and the extended thinking of Claude Haiku 4.5 is off by default (Table 16). Gemini 2.5 Pro has a 4,000-token budget. A response with the wrong number of items or an out-of-range value counts as a failure and is retried. Table 3 lists the models used in each analysis.

System message.

> You are participating in a clinical training simulation. You will play the role of a person completing a routine mental-health screening questionnaire. Answer every item in character. Respond only with the requested JSON.

Persona line. For severity "moderately severe":

> **label:** You are a person living with moderately severe depression. Answer the following questionnaire as that person would answer it about themselves.
>
> **impairment:** You are a person who has been having a hard time over the past two weeks. You have missed several days of work, most household tasks have gone undone, and you have not been returning messages from friends. Answer the following questionnaire as that person would answer it about themselves.
>
> **chart:** You are a patient at a mental health clinic, here for a routine assessment. Your chart records a diagnosis of moderately severe depression. Answer the following questionnaire as you would at that appointment.
>
> **honest control:** Answer the following questionnaire truthfully about yourself, as an ordinary adult with no particular mental-health condition.

The impairment texts describe role functioning only and mention no symptom. The coached arm appends: "Be aware that questionnaires like this include statistical checks that detect exaggerated or inconsistent answering. Answer in a way that is realistic and internally consistent, and do not overstate your symptoms." The questionnaire block gives the instrument's instruction, its response options and the items in a per-trial random order, and asks for `{"a1": n, ..., "aK": n}`. For the extension instruments the condition in the persona line is "psychological distress", "obsessive-compulsive disorder", "somatic symptom disorder", "post-traumatic stress related to your experience of the COVID-19 pandemic", "eating disorder symptoms" or "post-traumatic stress disorder

(PTSD)"; the EDE-Q is shown in its four sections, each with its own response scale, and items are shuffled within sections.

Feigning study roles. All three roles begin "You are a university student, about 21 years old." *Patient:* in the last two years you had a psychological (or physical) injury that made it impossible to meet your duties for at least a month and you still have symptoms; answer honestly. *Feigner:* you have not had any injury, but the university offers a tuition deduction to students who had one; you are not eligible, want the deduction, and no one would ever find out; complete the questionnaire as if you had such an injury, in a way that would get you the deduction. *Honest:* you have not had any injury; answer honestly. Injury type is drawn to match the human groups (68% and 57% psychological); 60 trials per role.

Interventions. *People differ*: "Keep in mind that people with the same level of depression differ a lot from one another. Most have some of the problems below not at all, a few quite strongly, and the rest somewhere in between. Answer as one specific person, not as an average." *Biography*: a name, age, occupation and life stressor drawn from fixed lists, with no symptom content, plus the severity label. *Numeric*: "You are a person whose total score on the following questionnaire would be about t out of 27", t drawn inside the band, no severity word. *Few-shot*: five real NHANES answer sheets from the target band, drawn only from respondents with even sequence numbers; evaluation uses the odd half.

Published personas. The persona line is replaced by "Your character is the following patient:" and a TalkDep persona (Wang et al., 2025), or by "Your character is the client described in this counselling intake form:" and a CACTUS intake form (Lee et al., 2024), followed by "Answer the following questionnaire as this person would answer it about themselves." and the same questionnaire block; the system message is unchanged. Each TalkDep persona (300–580 words, written with three clinical psychologists) gives a name, age, gender, BDI-II score and severity, four key symptoms with a quote and a severity each, life history, social context, communication style and behavioural constraints. Each CACTUS intake form, used there to drive simulated counselling clients, gives demographics, the presenting problem, the reason for seeking counselling, past history, functioning and relationships; we sample 400 distinct forms with a fixed seed. In the case-formulation variant the lead is "Your character is the client described in this counselling intake form, with the automatic thought and cognitive distortions from their case formulation:" and the form is followed by the automatic thought and the cognitive distortions that CACTUS records with that client (its first record); this is a small part of a CBT cognitive model such as those of PATIENT-ψ (Wang et al., 2024b), which also describe core beliefs, intermediate beliefs and coping strategies and whose data are released on request. The GAD-7 is given with the same personas and its own questionnaire block. The persona runs added for all 17 models used the Azure deployments of GPT-4o-mini, GPT-5.2 and GPT-5.4-mini, and reasoning is off as in the main runs. At totals 15–24, items 1–8 are answered "not at all" 0–4% (TalkDep) and 0–3% (CACTUS) of the time, against 9.9% for real respondents, and in all 204 TalkDep model–persona pairs more symptoms are endorsed than by real respondents with the same totals. For the share of distinct answer patterns, real respondents are drawn at the totals of a model's responses with total at least 5, one per response and without replacement within a total (50 draws), so two people may give the same pattern. The personas mainly lower the rate of suicidal ideation, from 84–100% with the one-line persona to 32–72% with TalkDep and 0–20% with CACTUS (real: 42%). For Table 22 we map each TalkDep key symptom to the PHQ-9 item it describes (e.g. sadness to item 2, fatigue to item 4); symptoms without a PHQ-9 counterpart, such as irritability, are left out. With the three mildest personas (PHQ-9 3–5) the models answer the listed symptoms and few others, 99–100% of endorsed answers at "several days" against 78–83% for real respondents with the same totals, so they endorse unlisted symptoms less often than real respondents (7–33% against 27–36%) but more symptoms in total (3.3–4.7 against 2.6–3.6). From moderate severity on they endorse unlisted symptoms as often or more often (54–100% against 55–93%).

Log-probability probe. Each item is asked alone after the persona line: "Over the last 2 weeks, how often have you been bothered by the following problem? … Reply with only a single digit: 0, 1, 2 or 3." We read $P(0), \ldots, P(3)$ from the top log-probabilities (at most 5 for GPT-5.x, 8 for Grok, 20

otherwise; four candidates fit every limit), renormalised. GPT-5-mini, Claude Haiku 4.5, Mistral-Large-3, Phi-4 and the Gemini models return no log-probabilities. Kimi-K2.5 is queried with reasoning off, as in the questionnaire runs; with reasoning on, the candidates listed after its reasoning block did not match the sampled tokens. GPT-5.2 and GPT-5.4-mini generate no reasoning tokens at their API default (Table 16), so their distributions are not conditioned on hidden reasoning. The whole-form entropy in Table 15 uses the 50 naive-arm forms of each severity cell; for real respondents, 50 are drawn from each band 200 times, so both use the same plug-in estimator at the same sample size.

Interviews. The patient model receives: "You are role-playing a patient in a clinical intake interview, as part of a clinician-training simulation. Stay in character throughout. Answer each question naturally, in one to three sentences, as this person would speak. Do not mention questionnaires, scores or ratings." followed by either the severity persona (free) or "over the last two weeks this person has experienced the following (private background; never recite it as a list)" and the nine symptoms with their frequencies (oracle; 20 NHANES profiles sampled per severity band), or a profile of the same form sampled from a graded response model (sampled; 20 per band), or "Your character is the following patient:" and a TalkDep persona in full, as in the questionnaire runs (seven interviews per persona, 84 per patient model). The model is fit by marginal maximum likelihood to NHANES 2005–2012 only (N=20,685); profiles are drawn with $\theta \sim N(0, 1)$ and items independent given θ, and kept if their total falls in the band. Table 30 compares profiles drawn the same way, but selected to match the total distribution of the 2013–2023 respondents, with those respondents. The interviewer opens and asks nine fixed questions, one per PHQ-9 domain, e.g. "How has your energy been during the day?" and "Some people who feel low have thoughts that they would be better off dead, or of hurting themselves. Have you had thoughts like that?". The rater ("You are an experienced clinician. Read the interview transcript and rate the patient on each PHQ-9 item for the last two weeks, using only what the patient said.") returns nine integers. Gemini 2.5 Pro patients get a 3,000-token budget per turn for thinking. Figure 10b shows that from moderate severity on, 95–100% of severity-persona interviews disclose thoughts of being better off dead, and Table 35 that the agent rates 34–51% of patients given a real profile above their band.

Human ratings. We drew 64 interviews without a filtered answer: 40 with a real profile (two per severity band for each of the five patient models), 10 with a GRM-sampled profile and 14 with a severity persona (two per patient model). Raters saw only the transcripts, under neutral identifiers and in an order set by their rater code, in an offline form that gave the PHQ-9 items, the four answer options with the usual frequency words ("some days" for several days, "most days" for more than half the days, "every day" for nearly every day), the instruction to rate the last two weeks from what the patient said, and an option to mark a rating as unsure. Five graduate students without clinical or psychological training volunteered without payment and rated all 64 independently, in about an hour each; the first page of the form told them that the patients are played by AI models, that some transcripts mention thoughts of death or self-harm, and that they could take breaks or stop at any time. Agreement is computed on items 1–8 against the profile given to the patient model; the model rater is the one used in the paper, on the same transcripts, and intervals come from 2,000 bootstrap draws of transcripts (Table 29). Raters marked 22% of their ratings as unsure, and 35% of those for symptoms given as "several days".

C.1 Content filtering

The cloud platform's filter (self-harm category) blocked 75 of 27,785 prompts in the experiments before the extension (0.27%): 13–14 of the 50 coached-mild PHQ-9 prompts for each of the five OpenAI models whose PHQ-9 runs used the platform (68 in total), and one PCL-5 feigner prompt for seven of the eight OpenAI models. On the EDE-Q it blocked 53–58 of 450 prompts for each OpenAI model (self-harm category, prompt filter; up to about 20% in the moderately severe and severe cells); the other models, including the Gemini models on Google's platform, were not filtered, nor was any published-persona prompt. Every block came from the prompt filter, before any answer was generated, and the prompts of a cell differ only in item order, which does not shift answers (Figure 12c); blocking therefore cannot select on the answers, but it lowers the weight of the blocked cells (coached mild on the PHQ-9, the more se-

vere cells on the EDE-Q) among the responses at a given total. In interviews the filter removed Llama-3.3's answer to the suicidality question in 78.8% of free interviews. A direct test showed it also blocks denials that restate the question ("No, I have never had thoughts that I would be better off dead or of hurting myself": 4 of 4 blocked) and passes only denials that do not name the content (0 of 4 blocked), so Llama's suicidality cannot be observed, and we report only its items 1–8. The filter also removed 12 answers to items 1–8 in 9 free interviews, 6 answers in 6 real-profile interviews, 8 in 8 sampled-profile interviews and 16 in 12 TalkDep interviews; these interviews are treated as missing and excluded (71, 74, 72 and 72 remain). No answers of the other six patient models were removed. Including them, with the rater inferring the missing items from the rest of the conversation, raises Llama's free-interview diversity from 0.48–0.82 to 0.91–1.16; in the real-profile condition breadth changes by at most 0.04, the share endorsing all eight items by at most 1.1 points, and diversity by at most 0.29.

D Measures and Statistics

Bands. For the PHQ-9 and GAD-7 profile analyses we use totals 15–24 and 12–19, which contain enough responses from real respondents (n=1,402 and 1,188) and from every model. Severity-band analyses (Table 6) use the PHQ-9 cut points 0–4, 5–9, 10–14, 15–19, 20–27 for simulated and real responses alike. Within a band or in the 15–24 range, a model's totals are distributed differently from the real respondents' (in 15–24 the real mean is 18.0, the model means 18.2–20.5), so the real reference for each model is the expectation over real respondents with exactly the same totals, weighted by that model's totals; a model enters a band when it has at least 20 responses there. Diversity is averaged over the totals that have at least eight responses, using at most 400 responses per total drawn without replacement; the real reference for a model is averaged over the same totals. Two analyses have fewer responses per condition and use a wider range: the interventions (160 trials per condition) use totals 5–27, and the interviews (at most 80 per condition) use the total of items 1–8 with at least three interviews per total.

Total-matched protocol (Table 1). Simulated responses come from the naive and coached arms. Totals are binned exactly on short scales and in widths of 2, 3 and 4 on the OCI-R, PCL-5 and EDE-Q. Only simulated responses whose bin contains at least five real respondents are kept; the share kept is the coverage. Human values are the expectation over real respondents in the same bins, weighted by the model's bin counts; this equals 1:1 matching without the duplicate draws that sampling with replacement would create. Each model is compared at its own totals, so the values describe each model within its range of totals and do not rank the models. Diversity is the mean pairwise L_1 distance per item within a bin, averaged over bins where both groups have at least eight responses; a ratio is reported where at least five bins carry it, which excludes the postpartum PCL-5 (one to four bins per model) and one model on the PHQ-15 (Table 11). Separability uses balanced samples drawn without replacement within each bin, 20 draws, with a label-shuffled run of the same pipeline as the chance reference (0.46–0.52). The OSF PCL-5 reference is not included in this table: its 69 genuine patients are spread over totals 0–76, so few totals have the five real respondents the protocol requires; it is analysed with nearest-total matching in the feigning comparison. Tables 10 and 11 give each model's separability and diversity ratio, and Figure 15a–c shows item endorsement at matched totals on the OCI-R, PHQ-15 and EDE-Q. COACH patients endorse all nine items less often than NHANES respondents with the same totals (2.5% against 5.0% at COACH's totals). COACH and NHANES respondents are themselves separable (AUC 0.78; different country, age and mode), so for the clinical references the profile measures are more informative than AUC; Table 1 weights the real references by each model's totals, the COACH–NHANES comparison by COACH's own.

Separability uncertainty. Separability is not bootstrapped, since duplicated rows on both sides of a cross-validation split inflate the AUC; a real respondent matched to several simulated responses is likewise kept in one fold. For the feigning study we use 80% subsampling without replacement of both groups (200 draws for people, 60 for models) with fresh matching and folds each draw. The permutation test shuffles the labels of a group and the genuine patients and recomputes the same statistic, the mean over 50 matchings and fold splits (1,000 permutations; Holm adjustment over the 17 models). Nearest-total matching draws genuine patients with replacement (each model's 60 responses use

14–24 of the 69, one of them up to 18 times), so a patient drawn more than once is kept in one cross-validation fold. To check the scale of the test at the models' totals, we split the genuine patients 200 times into random halves, draw patients from one half at each model's totals and compare them with the other half: this gives 0.38–0.59 on average, against 0.62–0.92 for the models in the same splits. Uninjured honest students give an AUC below 0.5 because their totals lie mostly below the genuine range (median 11.5; five at 0, while no genuine patient is at 0 and five are at 5 or below), so matching reuses a few low-scoring patients and the folds reverse the prediction. The log-odds comparison fits, on the genuine patients, a logistic model of answering "not at all" with item effects and a cubic spline in the total, and estimates each group's shift from it; responses outside the genuine range of totals are left out, intervals and the paired difference from the feigners come from 1,000 bootstrap draws that resample the genuine patients, the group and the feigners together, and the in-range version reweights the models to the feigners' totals in bins of five points (Table 13). The halving test compares, in each of the 200 halvings, a group's AUC against one half with that of genuine patients drawn from the other half at the group's totals; p is the share of halvings in which the group is not more separable (unadjusted $p<0.05$ for 2 models, none after Holm adjustment). The direct test against the feigners shuffles the labels of model and feigner responses and recomputes the difference between their AUCs against the genuine patients (mean over 10 matchings, 1,000 permutations). Subsampling intervals (here and in Figure 9d) give the 2.5th and 97.5th percentiles over 80% subsamples; they show the stability of the statistic and are not calibrated confidence intervals.

Checks on the feigning comparison. With nearest-total matching the simulated and the matched genuine totals differ by 0.2–1.0 points on average (43–79% match exactly). The "not at all" ratio divides a group's share of "not at all" answers by that of genuine patients matched to its totals (mean over 50 matchings). Its intervals come from 4,000 bootstrap draws that resample the group, the feigners and the genuine patients together, so the difference from the feigners is paired; with Bonferroni adjustment over the 17 models (99.7% intervals) the feigners' ratio exceeds that of every model except Gemini 2.5 Flash by at least 0.09. Within the feigners' 10th–90th percentile of totals (6–51), leaving out Gemini 2.5 Flash with 4 responses there, the models' ratios are 0.00–0.43 against 0.87 for the feigners, and uninjured honest students give 0.99 (0.91–1.09). Feigners endorse 0.6 more symptoms than matched genuine patients (-0.2 to 1.6), the models 0.7–2.8 more (all lower bounds above zero). In points of "not at all", the models' gap exceeds the feigners' significantly for one model after Holm adjustment. Within the feigners' range of totals, which holds 11–60 responses per model apart from Gemini 2.5 Flash (Table 14), the models endorse 1.0–2.8 more symptoms (lower bounds at least 0.4) and the feigners 0.7 (-0.2 to 1.9). For the 15 models that raise both total and "extremely" answers when feigning, the 95% bootstrap intervals of both changes lie above zero (lower bounds 0.5–25.4 points of total and 3.8–75.8 points of "extremely"). Item scores divided by the total separate the models as well as raw scores (0.48–0.96; people told to feign 0.49), and restricting the models to totals that some genuine patient has gives 0.50–0.91. The genuine totals have median 33 (IQR 18–51, maximum 76). Figure 13c shows item endorsement in each group. Figure 7c shows that the over-reporting index flags no simulated patient from 14 of 17 models: it counts extreme answers on rare items, whereas the models differ in breadth. In the feigning role GPT-4o lowers its total and "extremely" answers slightly, and GPT-5.4-mini answered as an uninjured student in 22 of 59 trials.

Real-sample baselines. Under the separability test used for the models (1:1 total matching, 250 responses, 10 draws), NHIS PHQ-8 against NHANES items 1–8 at totals 12–21 gives 0.56 (s.d. 0.03), NHANES 2017–2023 against 2005–2010 at totals 15–24 gives 0.54 (0.03), and random halves of NHANES give 0.49 (0.04). Under the nine-reference protocol, two NHIS years of the GAD-7 (2022 against 2019) and of the K6 (2016 against 2018), the two pooled EDE-Q studies and two NHANES periods give 0.49–0.51 (Table 12). On the OCI-R, the postpartum PCL-5 and the EDE-Q, whose totals are binned, the exact total alone gives 0.39–0.46 within the same bin samples, and removing from every response the mean item scores of real respondents with its total leaves the item AUC unchanged (0.74–0.99), so differences in total within a bin do not drive separability. A classifier with one weight per answer level sepa-

rates the models almost perfectly on the PHQ-9 and GAD-7 (0.94–0.99) but not the feigners (0.33) or the GRM-sampled profiles (0.49), so the linear test is conservative.

Robustness. For GPT-4o-mini, asking one item per call gives the same breadth as answering the whole form, and sampling temperature has little effect (Figure 12a). Answers do not drift with item position (slope -0.006 per position; Figure 12c). Totals rise with the instructed severity (Figure 12b; calibration per model in Table 4).

Coaching. Telling models that questionnaires contain checks for exaggeration lowers breadth, averaged over the four severities, by 0.03–0.37 symptoms in every model.

Reasoning effort. The main runs have reasoning off, and Gemini 2.5 Pro thinks with its minimum budget (68 thinking tokens per PHQ-9 form; Table 16). Four models reason at their API default: GPT-5-mini (466 reasoning tokens per PHQ-9 form), Gemini 2.5 Flash (749), Gemini 2.5 Pro (1,304) and Kimi-K2.5 (1,760 generated tokens, not itemised). For GPT-5-mini (efforts minimal, low, medium and high) and Gemini 2.5 Flash (thinking off, low, dynamic and high) we repeated every experiment they took part in, for Gemini 2.5 Pro every experiment at its default dynamic thinking, and for Kimi-K2.5 the questionnaires with reasoning on (Tables 17–19); these settings add 37,790 questionnaire administrations and 960 interviews. Reasoning changes calibration and the first-answer correlation: without it GPT-5-mini, Kimi-K2.5 and Gemini 2.5 Pro place 52%, 57% and 49% of PHQ-9 totals in the instructed band and their first answer correlates with the rest of the form (r=0.31, 0.54 and 0.34), with it 87–99% and lower ($r \leq 0.15$), whereas for Gemini 2.5 Flash it does at every setting (r=0.35–0.60). In the feigning role, GPT-5-mini raises its total without reasoning ($+28.6$ over the patient role) but answers as an uninjured student with it (-37.1 to -55.2). The pattern persists at every setting, although diversity changes: all nine PHQ-9 items are endorsed in 87–99% and all seven GAD-7 items in 99–100% of responses, diversity is 1.14–2.15 on the PHQ-9 and 0.67–2.28 on the GAD-7, and the models are separable from real respondents on all nine references (AUC 0.60–0.99; real-to-model diversity ratio 1.8–8.2). In the feigning study the models answer "not at all" 0.00–0.37 times as often as genuine patients with the same totals at every setting (feigners 0.90) and are separable from them (AUC 0.63–0.84, permutation $p<0.05$); the ratio differs from the feigners' at every setting except Gemini 2.5 Flash without thinking (interval up to 0.44), whose totals (median 66) lie where only seven genuine patients score 60 or more. In the Gemini experiments five real answer sheets lower separability at every Gemini 2.5 Flash setting (0.62–0.75 from 0.79–0.83) but not for Gemini 2.5 Pro, already the least separable model with the one-line persona (0.64); the published personas remain separable (0.80–0.91), and interviews with a severity persona stay uniform (all eight items in 99–100% of interviews against 74–83% of real respondents with the same totals; diversity 0.27–0.83 against 4.35–5.94), while given profiles are recovered (item correlation 0.79–0.90; suicidal ideation in every interview).

Person fit and probes. Models keep the human ordering of items only in part: the Spearman correlation of model and human item means is 0.43–0.80 on the PHQ-9 and 0.29–0.79 on the GAD-7 (Figure 11c). We fit a graded response model by marginal maximum likelihood to NHANES (PHQ-9) and NHIS (GAD-7). l_z and normalised Guttman errors are computed against these parameters and compared with total-matched real respondents as Cohen's d. Every model makes fewer Guttman errors than total-matched real respondents (d -0.20 to -0.82), l_z is higher for 16 of 17 models on each instrument (the exception is GPT-4.1-nano), and the extreme-severity probe is lower for every model. The rare-symptom probe stays within d of ± 0.4; the rare-combination probe is raised for some models, most for GPT-4.1-nano (d 0.94 on the PHQ-9). The three probes are the surprisal of each item level given the total (rare symptoms), of pairs of items at ≥ 2 given the total (rare combinations), and the share of endorsed items at the top category (extreme severity), all estimated from the human reference. Applied to responses generated from the fitted model itself, these measures separate at AUC 0.48–0.55, so they do not flag well-calibrated responses.

Screening experiment. The task is to predict whether PHQ-9 item 9 is endorsed from items 1–8 with a logistic regression; it measures how simulated profiles relate item 9 to the other items and does not predict suicide risk. All real training data come from NHANES 2005–2012 (N=20,685), the cycles the GRM is fit on, and the test set is every

2013–2023 respondent, so no test respondent informs a training source; metrics use the 5,747 test respondents with an items 1–8 total of at least 5. 1,000 2005–2012 respondents with such totals are held out for recalibration (logistic regression of the outcome on the logit of the prediction) and never enter a training set. Each simulated training set (all nine cells of the main design) is compared with real training sets of the same size drawn from the remaining 2005–2012 respondents in three ways: at random; with the same items 1–8 total as each simulated response; and with the same total and the same item-9 answer, so that only the composition of items 1–8 differs (nearest total when none matches; 50 draws each). Intervals for the differences come from 1,000 paired bootstrap replicates that resample the simulated training set and the test set and draw fresh real controls. The calibration slope regresses the outcome on the logit of the prediction.

Screening results. Figure 16 and Tables 31–34 give the results. Trained on simulated patients, the model predicts near-zero risk below a total of 10 and 73–99% from 15, where 38.5–57.5% is observed (Figure 16a), and its calibration slope is 0.17–0.27 against 0.83 for the real-trained model. Against real training sets of the same size drawn at random, AUC is similar (0.72–0.81 against 0.76–0.78); against real sets with the same items 1–8 totals (0.80–0.81) it is lower for 16 of 17 models, significantly for 5 (pooled -0.027, 95% interval -0.040 to -0.015; Figure 16b). Matching on item 9 as well removes the difference (pooled $+0.008$, -0.003 to 0.018), so it comes from how often simulated patients endorse item 9 at a given severity. Recalibrating on 1,000 held-out real respondents brings the Brier score to 0.097–0.106 against 0.098–0.100 for the matched real sets, without changing AUC (Figure 16c), and GRM-sampled profiles give a model calibrated like the real-trained one (AUC 0.79 against 0.82). Added to a small real training set, simulated patients raise AUC by 0.02–0.05 and, from 100 real respondents on, worsen the Brier score.

Severity labels and totals. Table 38 gives each model's totals by instructed severity; in the coached arm the share above the band falls to a median of 51%. Told only how daily life is affected (no severity word), GPT-4o-mini and GPT-5.2 give median totals of 12 and 13 at the impact written for mild depression, above the mild band in 98% and 94% of responses. Given a target total instead of a severity word, the share in the instructed band rises from 25–48% to 28–100% (Table 20), and to 99–100% for GPT-4.1 and Gemini 2.5 Flash and Pro.

Assessment agent. The agent's output is the PHQ-9 that the model rater codes from an interview (GPT-4.1, or Claude Haiku 4.5 for GPT-4.1 patients). Its severity is the band of the rated total (none 0–4, mild 5–9, moderate 10–14, moderately severe 15–19, severe 20–27). The patient's severity is the one the patient was told, or the band of the profile it was given; profiles were drawn within each band, so the two agree by construction for real and sampled profiles. For symptoms 1–8 we also score presence: the share rated absent, the share of symptoms absent from the given profile that are rated absent, and the share whose presence or absence matches the profile. We do not score the most prominent symptom, because most rated and given profiles tie several symptoms at the highest level. For TalkDep personas, the patient's severity is the PHQ-9 band named like the persona's severity label: minimal (BDI-II 5–7) as none, mild (12–15), moderate (22–28), moderate to severe (40) as moderately severe, and severe (35–38); TalkDep's own evaluation ranks the personas by BDI-II score, so we also report the Spearman correlation of the rated total with it (Table 28). The human ratings of 64 transcripts are scored the same way (Table 35). Their 14 severity-persona transcripts include five severe ones, whose band is placed correctly most often, so the band rates are averaged with equal weight over severities, as in the full design; unweighted, human raters place 41% of patients told a severity in their band and the model rater 43%.

Residual structure. Within each total from 5 to 24, we subtract a group's mean item scores from each of its responses and correlate the residuals across the nine items. The contrast is the mean residual correlation within the somatic items (sleep, fatigue, appetite, concentration, psychomotor) and within the non-somatic items (interest, mood, worthlessness, suicidal thoughts) minus that between the two groups, the split that fits PHQ-9 data best among the two-factor models compared by Elhai et al. (2012); the similarity is the correlation of a group's 36 residual correlations with those of all NHANES respondents. Real respondents drawn at each model's totals, one per response, give the reference for a sample of the model's size (100 draws); a model's contrast is weaker or stronger

than real when its 95% bootstrap interval lies below or above the mean of these draws. Because the residuals of a response sum to zero, residual correlations depend on how much each item varies at a given total as well as on how items covary. The structure-free reference therefore draws, one per model response and at its total, from the maximum-entropy distribution over the responses with that total whose item distributions equal the model's there, fitted by iterative proportional fitting (100 draws); it has the model's item distributions and no covariation beyond the fixed total. For the noise control, every item of a model's responses moves one level up or down with probability p, and p is set so that diversity matches that of real respondents with the same totals. A graded response model has one trait and treats items as independent given it, so its profiles show no somatic–non-somatic contrast (Table 37), although their diversity and separability match those of real respondents (Table 30). Diversity at a given total depends only on each item's answer distribution at that total, and the classifiers, including the one-hot classifier in Table 12, add up evidence item by item, so none of them detects a difference confined to how items covary. The five patient models in the interviews were given the same 80 profiles per condition, so we pool their interviews, compare the PHQ-9 rated from them (primary rater) with 80 real respondents drawn at the given profiles' totals (200 draws), and resample profiles for intervals; 80 real respondents agree with NHANES at 0.26–0.72; the profiles given agree at 0.34 (real) and 0.01 (GRM-sampled) and the answers rated from their interviews at 0.58 and 0.26, so the interview adds covariation of its own and the test cannot show whether a profile's structure is kept. Among the 36 residual correlations, all 17 models give a lower one than real respondents at their totals between sleep problems and fatigue (median -0.03 against 0.08), 15 a lower one between low mood and suicidal thoughts, and all 17 a higher one between poor concentration and low mood or loss of interest (-0.04 and -0.03 against -0.22 and -0.18). In real respondents with the same total, sleep problems come with fatigue and poor concentration trades off against the mood symptoms; the models weaken both.

E Proofs

For Propositions 1 and 2, let $X = (X_1, \dots, X_K)$ be a response drawn from a group of respondents with total t, with item distributions $P_1, \dots, P_K$, and let X' be an independent draw from the same group.

Proposition 1. By linearity of expectation,

$$D = \mathbb{E}\|X - X'\|_1 = \textstyle\sum_k \mathbb{E}\,|X_k - X'_k| \\ = \textstyle\sum_k \sum_{a,b} |a-b|\, P_k(a)\, P_k(b),$$

where the last step holds because X_k and X'_k are independent with distribution P_k. So D depends only on $P_1, \dots, P_K$. The same holds for our estimate, the mean of $\|x_i - x_j\|_1$ over pairs $i \neq j$ of responses with total t: it is the sum over items of the mean of $|x_{ik} - x_{jk}|$ over the same pairs, which depends only on how often each answer occurs on item k. Averaging over totals keeps this property. □

Proposition 2. Pool the two groups, with label $y \in \{0, 1\}$ and $\pi = \Pr(y = 1)$. Let $\phi(x)$ stack item-wise features $\phi_k(x_k)$, and let $\eta = \beta_0 + \beta^\top \phi(X)$. The logistic regression maximises the expected log-likelihood

$$\ell(\beta_0, \beta) = \mathbb{E}\big[\, y\,\eta - \log(1 + e^\eta)\big],$$

which is concave in (β_0, β). At $\beta = 0$ and $\beta_0 = \log \frac{\pi}{1-\pi}$, the gradient is $\mathbb{E}[y - \pi] = 0$ in β_0 and, in β,

$$\mathbb{E}\big[\phi(X)(y - \pi)\big] = \pi(1-\pi)\big(\mathbb{E}[\phi(X) \mid y{=}1] \\ - \mathbb{E}[\phi(X) \mid y{=}0]\big).$$

Each component of ϕ depends on one item, so its mean depends only on that item's distribution, which is the same in both groups; the gradient is zero. A stationary point of a concave function is a maximum, so the fitted model gives every response the same probability, and its AUC is $1/2$. With an L_2 penalty, as in our classifiers, the objective is strictly concave, has the same gradient at this point, and has a unique maximum. For total-matched samples the condition holds when the two groups have the same item distributions at each total. □

Proposition 3. Write $\hat{X}_{<k}$ for $(\hat{X}_1, \dots, \hat{X}_{k-1})$. By the definition of mutual information, $H(\hat{X}) = I(\Pi; \hat{X}) + H(\hat{X} \mid \Pi)$, and $I(\Pi; \hat{X}) \leq H(\Pi)$. For each persona π, the chain rule gives

$$H(\hat{X} \mid \Pi{=}\pi) = \textstyle\sum_k H(\hat{X}_k \mid \hat{X}_{<k}, \Pi{=}\pi) \\ \leq \textstyle\sum_k H(\hat{X}_k \mid \Pi{=}\pi) = h(\pi),$$

because conditioning does not increase entropy, with equality when the items are independent given π. Averaging over π gives $H(\hat{X} \mid \Pi) \leq \sum_k H(\hat{X}_k \mid \Pi)$, and with the first identity the inequality in the proposition. The perplexity of the answers to π is $2^{H(\hat{X}|\Pi=\pi)} \leq 2^{h(\pi)}$. For two independent answers to π, with p the probability of an answer, Jensen's inequality for the convex function $u \mapsto 2^u$ gives

$$\begin{aligned}\Pr(\hat{X} = \hat{X}') = \textstyle\sum_x p(x)^2 &= \mathbb{E}\big[2^{\log_2 p(\hat{X})}\big] \\ &\geq 2^{\mathbb{E}[\log_2 p(\hat{X})]} \\ &= 2^{-H(\hat{X}|\Pi=\pi)} \geq 2^{-h(\pi)}.\end{aligned}$$

□

Proposition 4. Write ℓ_k for $\ell_k(s)$ and let $e_k = q(\hat{X}_k \neq \ell_k \mid \pi) \leq \varepsilon$. By the union bound, $\Pr(\hat{X}_k = \ell_k \; \forall k) \geq 1 - \sum_k e_k \geq 1 - K\varepsilon$, and on this event $\hat{T} = \sum_k \ell_k$. If every $\ell_k \geq 1$, every item is endorsed on the same event, so $A \geq 1 - K\varepsilon$, and $\Pr(\hat{X}_k = 0) \leq e_k$ gives $Z \leq \varepsilon$. For one item, the grouping rule gives

$$\begin{aligned}H(\hat{X}_k \mid \pi) &= h_2(e_k) + e_k \, H(\hat{X}_k \mid \hat{X}_k \neq \ell_k, \pi) \\ &\leq h_2(e_k) + e_k \log_2 m,\end{aligned}$$

because $\hat{X}_k \neq \ell_k$ leaves m values. The bound increases in e_k for $e_k \leq \frac{1}{2}$, since its derivative is $\log_2 \frac{(1-e_k)m}{e_k} \geq 0$, so it is at most $h_2(\varepsilon) + \varepsilon \log_2 m$. Finally, $|\hat{X}_k - \hat{X}'_k| \leq m \, \mathbf{1}\{\hat{X}_k \neq \hat{X}'_k\}$ and

$$\Pr(\hat{X}_k \neq \hat{X}'_k) \leq 1 - (1 - e_k)^2 \leq 2e_k \leq 2\varepsilon,$$

so $\mathbb{E}\|\hat{X} - \hat{X}'\|_1 \leq 2mK\varepsilon$. □

In the main text we apply Proposition 3 at each model's item entropy averaged over the items and the four symptomatic severity cells (Table 15), which gives the geometric mean of the bounds of the single cells; per cell the bounds are 2–240 for the models and 14–51,829 for real respondents at the same totals. These plug-in estimates from 50 forms are biased downwards, so the bounds are approximate.

F Full Results

Analysis	n	Models
PHQ-9, GAD-7, PCL-5 roles, six extension instruments; person fit, first-answer correlation, screening	17	all
Log-probability probe	10	GPT-4o-mini, GPT-4o, GPT-4.1-nano, GPT-4.1-mini, GPT-4.1, GPT-5.2, GPT-5.4-mini, Grok-4.1, Llama-3.3, Kimi-K2.5
Interventions; interviews, told a severity	7	GPT-4o-mini, GPT-4.1, Haiku 4.5, Llama-3.3, Gemini-2.5-FL, Gemini-2.5-F, Gemini-2.5-Pro
Published personas in the intervention protocol (TalkDep, CACTUS, CACTUS with case formulation)	7	GPT-4o-mini, GPT-4.1, Haiku 4.5, Llama-3.3, Gemini-2.5-FL, Gemini-2.5-F, Gemini-2.5-Pro
Published personas, PHQ-9 and GAD-7 (the same three sets)	17	all
Interviews, given a real profile	5	GPT-4o-mini, Llama-3.3, Gemini-2.5-FL, Gemini-2.5-F, Gemini-2.5-Pro
Interviews, given a sampled profile	5	GPT-4o-mini, Llama-3.3, Gemini-2.5-FL, Gemini-2.5-F, Gemini-2.5-Pro
Scale ladders	10	GPT-4o-mini, GPT-4o, GPT-4.1-nano, GPT-4.1-mini, GPT-4.1, GPT-5-mini, GPT-5.2, Gemini-2.5-FL, Gemini-2.5-F, Gemini-2.5-Pro
Robustness (temperature, format, framing)	2	GPT-4o-mini, GPT-5.2
Reasoning effort (two to four settings, including the API default): questionnaires, feigning roles, nine references and screening; for the Gemini models also interventions, published personas and interviews	4	GPT-5-mini, Kimi-K2.5, Gemini-2.5-F, Gemini-2.5-Pro

Table 3: Models used in each analysis.

Model	n	Breadth	Real$_=$	All items (%)	Real$_=$	Not at all (%)	Real$_=$	Unev.	Div.	Real$_=$	ρ	AUC	All 1–8	Real$_=$	Calib.	Div. $\%_{5–27}$	SI
Real respondents	1402	7.64 [7.59, 7.69]	—	20.5 [18.5, 22.7]	—	15.1 [14.6, 15.7]	—	1.05	7.35	—	1.00	—	—	—	—	—	43
GPT-4o-mini	288	8.98 [8.96, 8.99]	7.76	97.9 [96.2, 99.3]	22.9	0.2 [0.1, 0.4]	13.8	0.66	1.95	7.76	0.48	0.86	100.0	47.9	42	23	98
GPT-4o	225	8.95 [8.92, 8.97]	7.89	94.7 [91.6, 97.3]	27.4	0.6 [0.3, 0.9]	12.3	0.63	2.43	7.35	0.63	0.78	100.0	53.8	59	29	95
GPT-4.1-nano	260	9.00 [8.99, 9.00]	7.75	99.6 [98.8, 100.0]	23.0	0.0 [0.0, 0.1]	13.8	0.44	2.32	7.35	0.43	0.85	100.0	48.4	47	32	100
GPT-4.1-mini	228	8.79 [8.74, 8.84]	7.94	79.4 [73.7, 84.2]	28.1	2.3 [1.8, 2.9]	11.7	0.69	1.86	7.35	0.72	0.79	100.0	54.6	49	23	79
GPT-4.1	253	8.96 [8.94, 8.98]	7.77	96.4 [93.7, 98.4]	24.4	0.4 [0.2, 0.7]	13.7	0.55	0.88	7.29	0.72	0.71	100.0	48.8	66	13	96
GPT-5-mini	198	8.87 [8.83, 8.91]	7.96	87.4 [82.8, 91.4]	29.5	1.4 [1.0, 1.9]	11.6	0.61	1.31	6.97	0.73	0.77	100.0	56.1	52	17	87
GPT-5.2	293	9.00 [8.99, 9.00]	7.95	99.7 [99.0, 100.0]	29.0	0.0 [0.0, 0.1]	11.6	0.66	0.83	7.11	0.80	0.81	100.0	55.6	51	10	100
GPT-5.4-mini	230	8.82 [8.77, 8.87]	7.99	82.2 [77.0, 86.5]	30.9	2.0 [1.5, 2.6]	11.3	0.64	1.40	7.35	0.72	0.76	100.0	58.7	51	16	82
Haiku 4.5	260	8.92 [8.88, 8.95]	7.91	91.5 [88.5, 94.6]	27.8	0.9 [0.6, 1.3]	12.1	0.68	0.97	7.50	0.72	0.85	100.0	55.8	42	14	92
Grok-4.1	235	8.59 [8.53, 8.66]	7.84	60.0 [53.6, 66.4]	25.9	4.5 [3.8, 5.2]	12.8	0.79	1.53	7.35	0.75	0.80	99.1	52.4	54	20	60
Llama-3.3	141	8.79 [8.72, 8.86]	7.75	79.4 [72.3, 85.8]	23.1	2.3 [1.6, 3.1]	13.9	0.64	1.58	8.04	0.50	0.74	100.0	48.7	44	16	79
Kimi-K2.5	189	8.89 [8.84, 8.93]	7.67	88.9 [84.1, 92.6]	22.1	1.2 [0.8, 1.8]	14.8	0.57	1.44	7.75	0.73	0.74	99.5	47.3	57	17	89
Mistral-L3	216	8.95 [8.92, 8.98]	7.88	94.9 [91.7, 97.7]	26.4	0.6 [0.3, 0.9]	12.4	0.67	1.28	7.35	0.75	0.76	100.0	52.6	48	18	95
Phi-4	307	8.67 [8.62, 8.72]	8.07	67.1 [61.9, 72.3]	31.9	3.7 [3.1, 4.2]	10.4	0.72	1.63	7.11	0.43	0.85	100.0	59.6	32	24	67
Gemini-2.5-FL	270	8.94 [8.91, 8.97]	7.85	94.4 [91.5, 97.0]	25.1	0.6 [0.3, 0.9]	12.8	0.73	2.06	7.35	0.57	0.86	99.3	51.5	48	27	95
Gemini-2.5-F	242	8.93 [8.90, 8.96]	7.99	93.4 [90.1, 96.3]	29.0	0.7 [0.4, 1.1]	11.2	0.68	1.75	7.35	0.73	0.79	100.0	56.0	42	20	93
Gemini-2.5-Pro	234	8.88 [8.85, 8.92]	7.84	88.5 [84.6, 92.3]	25.5	1.3 [0.9, 1.7]	12.9	0.70	1.83	7.35	0.80	0.68	99.6	51.2	49	21	89

Table 4: PHQ-9, responses with total 15–24, with 95% intervals. Breadth: symptoms endorsed (of 9). Unev.: within-person s.d. Div.: diversity, over the totals with at least eight responses. ρ: Spearman correlation with the human item ordering. Real$_=$: the preceding measure among real respondents at the model's exact totals. AUC: separability from total-matched real respondents (two real surveys: 0.56). All 1–8: % endorsing every one of items 1–8. Calib.: % of all trials whose total falls in the instructed band. Div. $\%_{5–27}$: diversity over totals 5–27 as a percentage of that of real respondents with the same totals (Figure 9e). SI: % endorsing item 9 at totals 15–24.

Model	n	Breadth	Real$_=$	All items (%)	Real$_=$	Not at all (%)	Real$_=$	Unev.	Div.	Real$_=$	ρ	AUC
Real respondents	1188	6.27 [6.22, 6.32]	—	47.0 [44.0, 50.2]	—	10.4 [9.7, 11.1]	—	0.95	5.78	—	1.00	—
GPT-4o-mini	318	7.00 [7.00, 7.00]	6.40	100.0 [100.0, 100.0]	51.2	0.0 [0.0, 0.0]	8.6	0.59	1.56	5.78	0.54	0.76
GPT-4o	267	7.00 [7.00, 7.00]	6.44	100.0 [100.0, 100.0]	56.4	0.0 [0.0, 0.0]	8.0	0.54	1.79	5.78	0.29	0.77
GPT-4.1-nano	265	7.00 [7.00, 7.00]	6.43	100.0 [100.0, 100.0]	54.9	0.0 [0.0, 0.0]	8.2	0.36	1.24	5.78	0.36	0.76
GPT-4.1-mini	238	7.00 [7.00, 7.00]	6.51	100.0 [100.0, 100.0]	59.0	0.0 [0.0, 0.0]	7.1	0.53	1.67	5.78	0.43	0.74
GPT-4.1	247	7.00 [7.00, 7.00]	6.30	100.0 [100.0, 100.0]	50.0	0.0 [0.0, 0.0]	9.9	0.46	1.39	5.78	0.57	0.73
GPT-5-mini	227	7.00 [7.00, 7.00]	6.47	100.0 [100.0, 100.0]	57.3	0.0 [0.0, 0.0]	7.6	0.48	1.14	5.70	0.43	0.76
GPT-5.2	235	7.00 [7.00, 7.00]	6.47	100.0 [100.0, 100.0]	56.8	0.0 [0.0, 0.0]	7.5	0.52	0.95	5.78	0.71	0.69
GPT-5.4-mini	257	7.00 [7.00, 7.00]	6.54	100.0 [100.0, 100.0]	60.7	0.0 [0.0, 0.0]	6.6	0.54	1.13	5.59	0.61	0.64
Haiku 4.5	193	7.00 [7.00, 7.00]	6.34	100.0 [100.0, 100.0]	51.1	0.0 [0.0, 0.0]	9.5	0.52	1.53	5.78	0.57	0.68
Grok-4.1	209	6.99 [6.98, 7.00]	6.36	99.0 [97.6, 100.0]	51.2	0.1 [0.0, 0.3]	9.2	0.63	1.21	5.78	0.75	0.64
Llama-3.3	166	7.00 [7.00, 7.00]	6.39	100.0 [100.0, 100.0]	51.4	0.0 [0.0, 0.0]	8.7	0.57	0.78	5.50	0.68	0.67
Kimi-K2.5	243	7.00 [7.00, 7.00]	6.21	100.0 [100.0, 100.0]	46.6	0.0 [0.0, 0.0]	11.3	0.48	1.45	5.78	0.54	0.71
Mistral-L3	224	7.00 [7.00, 7.00]	6.37	100.0 [100.0, 100.0]	51.3	0.0 [0.0, 0.0]	8.9	0.58	1.05	5.78	0.79	0.64
Phi-4	277	7.00 [7.00, 7.00]	6.54	100.0 [100.0, 100.0]	61.2	0.0 [0.0, 0.0]	6.6	0.52	1.46	5.78	0.64	0.71
Gemini-2.5-FL	280	7.00 [7.00, 7.00]	6.35	100.0 [100.0, 100.0]	49.6	0.0 [0.0, 0.0]	9.2	0.60	1.67	5.78	0.39	0.72
Gemini-2.5-F	256	7.00 [7.00, 7.00]	6.55	100.0 [100.0, 100.0]	61.8	0.0 [0.0, 0.0]	6.5	0.56	1.87	5.78	0.64	0.63
Gemini-2.5-Pro	237	7.00 [7.00, 7.00]	6.41	100.0 [100.0, 100.0]	53.2	0.0 [0.0, 0.0]	8.4	0.61	1.65	5.78	0.79	0.68

Table 5: GAD-7, responses with total 12–19, with 95% intervals. Breadth: symptoms endorsed (of 7). Unev.: within-person s.d. Div.: diversity, over the totals with at least eight responses. ρ: Spearman correlation with the human item ordering. Real$_=$: the preceding measure among real respondents at the model's exact totals. AUC: separability from total-matched real respondents (two NHIS years: 0.49, Table 12).

Band	Models	Breadth	Gap	N.a.a. (%)	Unev.	Div.
0-4 (none)	16	1.55 / 1.81	+0.28	82.8 / 79.9	0.38 / 0.33	2.23 / 0.40
5-9 (mild)	7	5.12 / 6.60	+1.45	43.1 / 26.7	0.89 / 0.61	7.57 / 1.11
10-14 (moderate)	17	6.58 / 7.97	+1.32	26.9 / 11.4	1.05 / 0.68	8.73 / 1.30
15-19 (mod.-severe)	17	7.49 / 8.84	+1.29	16.7 / 1.8	1.09 / 0.69	8.76 / 1.87
20-27 (severe)	17	8.42 / 9.00	+0.58	6.4 / 0.0	0.81 / 0.56	5.46 / 1.12

Table 6: PHQ-9 by total-score band. Each cell: real respondents at the models' totals / models, median over models with at least 20 responses in the band. Gap: median over models of each model's breadth minus that of real respondents at its totals.

Model	l_z	G^p	Rare	Comb.	Extr.
PHQ-9					
GPT-4o-mini	+0.03	-0.24	+0.24	+0.32	-1.01
GPT-4o	+0.17	-0.29	+0.08	+0.26	-0.76
GPT-4.1-nano	-0.58	-0.33	+0.37	+0.94	-0.94
GPT-4.1-mini	+0.35	-0.45	-0.02	+0.16	-0.70
GPT-4.1	+0.25	-0.72	-0.01	+0.25	-0.84
GPT-5-mini	+0.28	-0.44	+0.03	+0.38	-0.80
GPT-5.2	+0.49	-0.68	-0.14	-0.07	-0.74
GPT-5.4-mini	+0.34	-0.44	-0.03	+0.27	-0.69
Haiku 4.5	+0.41	-0.54	-0.06	+0.05	-0.76
Grok-4.1	+0.69	-0.60	-0.17	-0.02	-0.64
Llama-3.3	+0.33	-0.60	-0.05	+0.40	-0.87
Kimi-K2.5	+0.29	-0.74	-0.07	+0.36	-0.94
Mistral-L3	+0.47	-0.60	-0.10	+0.02	-0.79
Phi-4	+0.13	-0.20	+0.06	+0.27	-0.60
Gemini-2.5-FL	+0.23	-0.25	+0.05	-0.06	-0.70
Gemini-2.5-F	+0.37	-0.50	-0.02	+0.06	-0.65
Gemini-2.5-Pro	+0.56	-0.75	-0.12	+0.04	-0.77
GAD-7					
GPT-4o-mini	+0.28	-0.67	+0.21	+0.20	-0.80
GPT-4o	+0.16	-0.21	+0.19	+0.32	-0.67
GPT-4.1-nano	-0.19	-0.44	+0.18	+0.67	-0.79
GPT-4.1-mini	+0.22	-0.35	+0.18	+0.26	-0.67
GPT-4.1	+0.17	-0.61	-0.03	+0.41	-0.70
GPT-5-mini	+0.26	-0.35	+0.10	+0.28	-0.65
GPT-5.2	+0.30	-0.71	+0.10	+0.21	-0.70
GPT-5.4-mini	+0.43	-0.57	+0.02	+0.20	-0.65
Haiku 4.5	+0.38	-0.57	-0.05	+0.13	-0.74
Grok-4.1	+0.58	-0.76	-0.12	-0.00	-0.62
Llama-3.3	+0.66	-0.76	-0.29	+0.03	-0.75
Kimi-K2.5	+0.29	-0.70	-0.07	+0.35	-0.82
Mistral-L3	+0.50	-0.82	-0.11	+0.04	-0.71
Phi-4	+0.14	-0.42	+0.20	+0.36	-0.63
Gemini-2.5-FL	+0.24	-0.28	+0.11	+0.21	-0.68
Gemini-2.5-F	+0.32	-0.35	+0.08	+0.16	-0.60
Gemini-2.5-Pro	+0.49	-0.73	-0.10	+0.09	-0.68

Table 7: Person-fit statistics and base-rate probes, Cohen's d against total-matched real respondents. Positive l_z and negative G^p: better fit than real responses. Rare, Comb., Extr.: rare-symptom, rare-combination and extreme-severity probes.

Item	a	b_1	b_2	b_3	%end.
little interest or pleasure in doi	2.33	0.80	1.69	2.20	25.8
feeling down, depressed, or hopele	3.15	0.75	1.61	2.07	25.3
trouble sleeping or sleeping too m	1.66	0.37	1.41	1.95	39.4
feeling tired or having little ene	2.01	-0.03	1.25	1.80	50.7
poor appetite or overeating	1.82	0.88	1.79	2.35	25.6
feeling bad about yourself, or tha	2.74	1.04	1.82	2.25	18.4
trouble concentrating on things	2.16	1.14	1.94	2.40	18.2
moving or speaking slowly, or bein	2.07	1.56	2.29	2.77	11.3
thoughts that you would be better	2.40	2.16	2.80	3.19	4.0

Table 8: Graded response model item parameters, PHQ-9, NHANES (N=41,714).

Item	a	b_1	b_2	b_3	%end.
feeling nervous, anxious, or on ed	3.54	0.77	1.61	1.93	24.0
not being able to stop or control	5.53	0.96	1.53	1.77	17.4
worrying too much about different	4.95	0.74	1.46	1.73	23.9
trouble relaxing	3.16	0.95	1.65	1.94	19.6
being so restless that it is hard	2.36	1.45	2.10	2.37	11.7
becoming easily annoyed or irritab	2.15	0.85	1.87	2.29	24.8
feeling afraid, as if something aw	2.79	1.32	2.02	2.32	12.4

Table 9: Graded response model item parameters, GAD-7, NHIS (N=31,132).

	PHQ-9 NHANES	PHQ-9 COACH†	GAD-7 NHIS	K6 NHIS	PHQ-15 C19PRC	OCI-R C19PRC	ITQ C19PRC	PCL-5 Postpartum†	EDE-Q Mexico
GPT-4o-mini	0.83	0.88	0.72	0.74	0.95	0.93	0.70	—	0.95
GPT-4o	0.77	0.83	0.76	0.60	0.87	0.92	0.63	0.91	0.88
GPT-4.1-nano	0.82	0.91	0.73	0.71	0.80	0.77	0.65	0.85	0.96
GPT-4.1-mini	0.76	0.73	0.71	0.66	0.88	0.91	0.69	0.95	0.92
GPT-4.1	0.68	0.86	0.71	0.64	0.86	0.93	0.63	0.91	0.88
GPT-5-mini	0.78	0.72	0.71	0.68	0.89	0.82	0.70	0.92	0.94
GPT-5.2	0.75	0.88	0.71	0.69	0.93	0.96	0.71	0.94	0.92
GPT-5.4-mini	0.75	0.77	0.65	0.64	0.94	0.98	0.79	0.94	0.91
Haiku 4.5	0.80	0.81	0.70	0.65	0.95	0.93	0.67	0.91	0.93
Grok-4.1	0.77	0.83	0.68	0.72	0.97	0.97	0.74	0.95	0.95
Llama-3.3	0.81	0.81	0.66	0.71	0.94	0.97	0.71	0.96	0.93
Kimi-K2.5	0.72	0.78	0.71	0.80	0.93	0.96	0.72	0.94	0.92
Mistral-L3	0.73	0.81	0.66	0.69	0.92	0.95	0.74	0.95	0.91
Phi-4	0.82	0.75	0.67	0.70	0.93	0.86	0.65	—	0.94
Gemini-2.5-FL	0.83	0.85	0.71	0.68	0.92	0.94	0.76	0.96	0.93
Gemini-2.5-F	0.79	0.80	0.67	0.64	0.90	0.93	0.64	0.95	0.91
Gemini-2.5-Pro	0.66	0.81	0.72	0.68	0.97	0.99	0.75	0.97	0.90

Table 10: Separability (AUC) from total-matched real respondents, per model and reference. †Clinical.

	PHQ-9 NHANES	PHQ-9 COACH†	GAD-7 NHIS	K6 NHIS	PHQ-15 C19PRC	OCI-R C19PRC	ITQ C19PRC	PCL-5 Postpartum†	EDE-Q Mexico
GPT-4o-mini	4.5 (13)	4.0 (12)	4.4 (11)	3.1 (15)	2.1 (8)	2.0 (9)	1.5 (13)	2.0 (1)	2.9 (8)
GPT-4o	3.5 (17)	3.2 (13)	3.9 (12)	3.8 (15)	2.4 (11)	2.3 (10)	1.5 (13)	2.5 (3)	3.0 (9)
GPT-4.1-nano	3.5 (17)	3.2 (12)	5.0 (14)	5.6 (14)	2.9 (11)	2.8 (13)	3.6 (12)	4.0 (4)	4.0 (14)
GPT-4.1-mini	4.7 (18)	4.4 (13)	5.2 (13)	5.1 (15)	3.3 (11)	3.2 (10)	3.3 (14)	5.3 (4)	3.8 (11)
GPT-4.1	8.6 (16)	8.0 (10)	5.4 (15)	4.3 (14)	5.1 (7)	4.0 (14)	2.5 (14)	5.1 (4)	5.0 (13)
GPT-5-mini	6.9 (15)	6.8 (11)	6.5 (13)	6.1 (13)	4.3 (11)	4.0 (10)	3.6 (13)	6.1 (3)	5.5 (12)
GPT-5.2	11.2 (14)	8.2 (10)	10.7 (14)	10.0 (12)	4.6 (8)	4.2 (13)	4.3 (12)	13.7 (3)	5.2 (9)
GPT-5.4-mini	7.4 (18)	6.1 (13)	10.1 (11)	8.1 (14)	5.4 (10)	3.2 (16)	4.1 (15)	6.8 (4)	4.6 (12)
Haiku 4.5	8.7 (14)	7.1 (13)	6.4 (12)	6.0 (12)	4.0 (9)	4.1 (13)	3.1 (11)	5.0 (4)	4.8 (15)
Grok-4.1	5.0 (17)	4.2 (13)	5.6 (16)	2.5 (16)	2.2 (9)	2.4 (13)	1.9 (15)	7.2 (3)	4.6 (14)
Llama-3.3	6.8 (13)	5.7 (11)	9.7 (13)	4.4 (13)	3.3 (10)	2.9 (16)	2.3 (13)	3.7 (4)	4.9 (6)
Kimi-K2.5	7.6 (16)	6.8 (12)	5.3 (15)	4.5 (14)	2.3 (6)	2.8 (14)	2.6 (11)	5.5 (4)	4.2 (13)
Mistral-L3	5.7 (16)	4.9 (13)	6.4 (13)	4.8 (12)	2.5 (8)	2.9 (13)	2.4 (13)	5.0 (4)	5.4 (10)
Phi-4	4.2 (14)	4.2 (9)	4.7 (12)	3.2 (14)	3.1 (3)	2.7 (8)	2.1 (12)	2.4 (1)	2.7 (9)
Gemini-2.5-FL	4.0 (16)	3.8 (13)	4.1 (11)	2.7 (17)	2.3 (9)	2.1 (11)	1.5 (11)	3.0 (2)	3.3 (15)
Gemini-2.5-F	5.4 (17)	4.8 (12)	4.1 (12)	3.7 (16)	3.1 (8)	2.0 (14)	2.1 (12)	4.8 (3)	4.5 (9)
Gemini-2.5-Pro	4.6 (18)	4.0 (13)	4.5 (14)	4.0 (15)	2.7 (8)	2.1 (13)	2.2 (12)	5.7 (2)	4.2 (8)

Table 11: Real-to-model diversity ratio at matched totals, per model and reference, with the number of totals behind each estimate in parentheses. †Clinical.

Reference	Compared	AUC
Two real samples, protocol of Table 1		
EDE-Q	study 1 vs study 2	0.51
GAD-7	NHIS 2022 vs 2019	0.49
K6	NHIS 2016 vs 2018	0.50
PHQ-9	NHANES 2017–2023 vs 2005–2010	0.51
Wide total bins: items / total alone / items, real item means at each total removed		
EDE-Q (width 4)	17 models	0.88–0.96 / 0.43–0.46 / 0.88–0.97
OCI-R (width 2)	17 models	0.76–0.99 / 0.43–0.46 / 0.74–0.98
PCL-5 postpartum (width 3)	17 models	0.85–0.97 / 0.39–0.43 / 0.82–0.99
Linear item scores / one-hot answer levels		
PHQ-9, totals 15–24	17 models	0.68–0.86 / 0.97–0.99
GAD-7, totals 12–19	17 models	0.63–0.77 / 0.94–0.97
PCL-5, OSF	people told to fake	0.53 / 0.33
PHQ-9, NHANES 2013–2023	GRM-sampled profiles	0.51 / 0.49

Table 12: Checks on separability. Two real samples: 400 respondents of the first sample against the second, five draws. Wide bins: within the same balanced bin samples as Table 1, the classifier given only the exact total, and given item scores minus the mean item scores of real respondents with the same total. One-hot: one weight per answer level of each item instead of one per item, same total matching and folds (feigners: 50 matchings; GRM: 600 profiles drawn at the totals of 2013–2023 respondents with totals $\geq$5).

		Breadth		"Not at all"									
Group	n	Group	Gen.$_=$	Group	Gen.$_=$	Diff.	Ratio [95% int.]	OR [95% int.]	PRI	AUC [sub.]	p	Δtotal	Δextr.
Genuinely injured	69	14.0	—	29.8	—	—	—	—	11.6	—	—	—	—
People told to fake	39	14.4	13.8	27.9	31.2	3.2	0.90 [0.75, 1.03]	0.71 [0.50, 0.95]	5.1	0.51 [0.31, 0.65]	0.35	—	—
Uninjured, honest	46	8.9	8.8	55.4	56.2	0.7	0.99 [0.91, 1.09]	—	2.2	0.29 [0.16, 0.57]	1.00	—	—
GPT-4o-mini	60	20.0	18.7	0.2	6.7	6.5	0.02 [0.00, 0.08]	0.03 [0.00, 0.07]	0.0	0.84 [0.60, 0.96]	<0.01	+3.8	+6.8
GPT-4o	60	20.0	19.2	0.1	4.2	4.1	0.02 [0.00, 0.08]	0.02 [0.00, 0.06]	11.7	0.63 [0.34, 0.88]	0.01	−2.4	−5.4
GPT-4.1-nano	60	19.9	18.6	0.7	7.2	6.5	0.09 [0.03, 0.18]	0.05 [0.02, 0.10]	0.0	0.46 [0.28, 0.88]	0.63	+3.7	+9.2
GPT-4.1-mini	60	20.0	18.8	0.1	5.9	5.8	0.01 [0.00, 0.05]	0.01 [0.00, 0.04]	0.0	0.75 [0.51, 0.94]	<0.01	+24.0	+71.1
GPT-4.1	60	20.0	18.6	0.0	6.9	6.9	0.00 [0.00, 0.00]	0 (none)	0.0	0.80 [0.55, 0.91]	<0.01	+9.8	+17.9
GPT-5-mini	60	20.0	18.9	0.2	5.7	5.6	0.03 [0.00, 0.07]	0.01 [0.00, 0.04]	0.0	0.66 [0.39, 0.96]	<0.01	+28.6	+80.8
GPT-5.2	60	19.9	18.2	0.2	8.8	8.5	0.03 [0.00, 0.07]	0.02 [0.00, 0.05]	0.0	0.70 [0.45, 0.83]	<0.01	+12.7	+34.0
GPT-5.4-mini	60	19.8	18.1	0.8	9.3	8.5	0.09 [0.03, 0.14]	0.05 [0.02, 0.09]	0.0	0.75 [0.47, 0.94]	<0.01	−14.1	+5.4
Haiku 4.5	60	19.4	16.6	3.2	17.2	14.1	0.18 [0.11, 0.28]	0.11 [0.07, 0.17]	0.0	0.62 [0.41, 0.90]	0.01	+18.8	+18.1
Grok-4.1	60	19.7	19.0	1.4	5.0	3.6	0.28 [0.14, 0.46]	0.19 [0.10, 0.31]	0.0	0.78 [0.43, 0.97]	<0.01	+18.1	+47.4
Llama-3.3	60	19.8	18.4	1.2	8.1	6.8	0.15 [0.07, 0.27]	0.10 [0.05, 0.17]	1.7	0.82 [0.66, 0.90]	<0.01	+12.0	+18.8
Kimi-K2.5	60	19.8	18.4	1.1	8.2	7.1	0.13 [0.05, 0.24]	0.11 [0.05, 0.19]	0.0	0.77 [0.59, 0.88]	<0.01	+12.0	+28.7
Mistral-L3	60	20.0	18.6	0.2	6.9	6.8	0.02 [0.00, 0.07]	0.02 [0.00, 0.05]	0.0	0.89 [0.64, 0.99]	<0.01	+8.0	+16.8
Phi-4	60	19.8	18.9	1.0	5.3	4.3	0.19 [0.09, 0.34]	0.14 [0.07, 0.25]	0.0	0.81 [0.44, 0.92]	<0.01	+7.4	+13.1
Gemini-2.5-FL	60	19.5	18.4	2.5	7.9	5.4	0.31 [0.18, 0.49]	0.22 [0.12, 0.33]	0.0	0.96 [0.87, 0.99]	<0.01	+23.3	+50.2
Gemini-2.5-F	60	20.0	19.2	0.2	4.1	3.9	0.04 [0.00, 0.44]	0.06 [0.00, 0.22]	16.7	0.77 [0.40, 0.90]	<0.01	+14.5	+53.1
Gemini-2.5-Pro	60	19.4	18.3	3.2	8.6	5.5	0.37 [0.22, 0.57]	0.26 [0.15, 0.40]	0.0	0.84 [0.59, 0.94]	<0.01	+14.3	+26.7

Table 13: PCL-5, OSF feigning study; model rows play a genuinely injured student. Gen.$_=$: genuinely injured respondents matched to the group's totals; Diff.: Gen.$_=$ minus group, in points; Ratio: group over Gen.$_=$, with 95% bootstrap interval. OR: odds ratio of answering "not at all" relative to genuinely injured respondents with the same total (logistic model with item effects and a spline in the total, fit on the genuinely injured; 95% bootstrap interval); every model's is below the feigners' after Holm adjustment over the 17 models. PRI: % flagged by the rare-items over-reporting index (≥ 3). AUC: separability from the genuinely injured group, with the 2.5th–97.5th percentiles over 80% subsamples and permutation p (every model with $p<0.05$ remains so after Holm adjustment over the 17 models). Δtotal, Δextr.: feigning minus patient role in total and in % "extremely" (people: -3.1 [-10.5, 3.7] and -6.8 [-12.5, -1.0]).

Group	n	N.a.a. diff.	Ratio	OR	Breadth diff.	p	p_{Holm}
People told to fake	31	3.6 [−0.8, 9.4]	0.87	0.70	0.72 [−0.16, 1.87]	—	—
GPT-4o-mini	14	7.1 [3.6, 11.1]	0.09	0.16	1.42 [0.71, 2.21]	0.13	0.86
GPT-4o	11	9.5 [4.5, 14.1]	0.05	0.006	1.90 [0.91, 2.82]	0.06	0.83
GPT-4.1-nano	56	6.7 [3.8, 11.8]	0.09	0.07	1.33 [0.75, 2.36]	0.10	0.86
GPT-4.1-mini	33	6.6 [3.2, 11.7]	0.02	0.007	1.31 [0.64, 2.33]	0.16	0.86
GPT-4.1	40	7.9 [4.6, 12.1]	0.00	0.000	1.57 [0.93, 2.43]	0.06	0.83
GPT-5-mini	56	5.6 [2.9, 10.8]	0.03	0.02	1.12 [0.57, 2.16]	0.17	0.86
GPT-5.2	31	10.9 [6.0, 15.0]	0.04	0.05	2.17 [1.19, 3.00]	0.02	0.22
GPT-5.4-mini	51	9.1 [6.4, 16.1]	0.10	0.06	1.82 [1.27, 3.22]	0.01	0.17
Haiku 4.5	60	14.2 [8.8, 19.5]	0.18	0.19	2.83 [1.77, 3.90]	<0.01	<0.01
Grok-4.1	16	8.3 [2.8, 15.0]	0.39	0.28	1.66 [0.56, 3.00]	0.09	0.86
Llama-3.3	38	6.8 [3.3, 12.9]	0.22	0.22	1.37 [0.66, 2.58]	0.10	0.86
Kimi-K2.5	42	8.3 [4.3, 12.1]	0.15	0.24	1.67 [0.86, 2.43]	0.07	0.83
Mistral-L3	24	7.5 [3.8, 11.9]	0.05	0.08	1.50 [0.75, 2.38]	0.09	0.86
Phi-4	21	5.7 [2.1, 10.7]	0.27	0.23	1.15 [0.43, 2.14]	0.22	0.86
Gemini-2.5-FL	38	5.4 [2.1, 10.1]	0.43	0.32	1.07 [0.42, 2.03]	0.23	0.86
Gemini-2.5-F	4	—	—	—	—	—	—
Gemini-2.5-Pro	52	5.2 [2.0, 9.2]	0.41	0.41	1.03 [0.40, 1.85]	0.25	0.86

Table 14: PCL-5 responses with totals in the feigners' 10th–90th percentile (6–51), against genuinely injured respondents matched to these totals, with 95% bootstrap intervals. N.a.a. diff.: matched genuine minus group, points of "not at all". Ratio: as in Table 13. OR: odds ratio as in Table 13, with the responses reweighted to the feigners' distribution of totals (bins of five points); every model's is below the feigners' after Holm adjustment. Breadth diff.: group minus matched genuine. p: one-sided bootstrap test that the model's "not at all" difference exceeds the feigners'; p_{Holm}: Holm-adjusted over the models shown; —, fewer than 10 responses in the range.

	Whole forms (17 models)									Log-probabilities (10 models)				
Model	H	Real$_=$	P_0	Real$_=$	Lock	Real$_=$	Dist.	Real$_=$	First r	H	P_0	Lock$_L$	Lock$_I$	Dist.
Real (bands)	1.47		0.285				72–100		—	1.51	0.285	44	—	72–100
GPT-4o-mini	0.59	1.51	0.007	0.183	44	48	19.6	91.8	0.27	0.19	0.000	74	81	3.2
GPT-4o	0.73	1.37	0.024	0.181	67	52	32.1	81.5	0.44	0.11	0.000	67	67	1.3
GPT-4.1-nano	0.70	1.30	0.000	0.149	85	52	28.8	76.2	0.57	0.35	0.000	70	70	8.2
GPT-4.1-mini	0.54	1.25	0.044	0.146	59	52	15.3	73.8	0.45	0.30	0.000	74	78	5.7
GPT-4.1	0.37	1.31	0.002	0.189	74	52	7.2	78.4	0.51	0.09	0.000	78	70	1.2
GPT-5-mini	0.45	1.33	0.091	0.213	70	56	11.2	78.0	0.31	—	—	—	—	—
GPT-5.2	0.29	1.32	0.000	0.165	56	48	4.4	78.2	0.36	0.10	0.000	67	56	0.9
GPT-5.4-mini	0.42	1.28	0.067	0.169	63	52	9.6	73.7	0.36	0.27	0.000	67	59	4.5
Haiku 4.5	0.41	1.30	0.004	0.136	52	48	9.0	77.7	0.62	—	—	—	—	—
Grok-4.1	0.51	1.37	0.107	0.189	52	52	15.2	80.4	0.53	0.21	0.069	85	81	3.3
Llama-3.3	0.43	1.30	0.096	0.199	63	52	11.8	75.8	0.58	0.03	0.000	63	70	0.5
Kimi-K2.5	0.49	1.44	0.058	0.203	59	56	13.4	82.8	0.54	0.44	0.016	85	59	12.9
Mistral-L3	0.46	1.28	0.000	0.144	59	52	10.6	74.4	0.58	—	—	—	—	—
Phi-4	0.56	1.15	0.024	0.101	52	33	17.9	69.3	0.41	—	—	—	—	—
Gemini-2.5-FL	0.67	1.38	0.000	0.134	44	48	24.7	83.2	0.47	—	—	—	—	—
Gemini-2.5-F	0.53	1.24	0.002	0.139	63	52	15.3	74.1	0.35	—	—	—	—	—
Gemini-2.5-Pro	0.53	1.32	0.040	0.178	63	52	16.7	77.8	0.34	—	—	—	—	—

Table 15: Per-item answer distributions, estimated from the answers across the 50 whole-form responses of each severity cell (all models) and from log-probabilities. Real$_=$: real respondents with the same totals as the model's forms in each cell (one per form, 200 draws). Real (bands): 50 real respondents per score band (whole forms) or the whole band (log-probabilities). H: entropy in bits, mean over items and the four symptomatic severities. P_0: share or probability of "not at all" at moderate severity. Dist.: % distinct among 400 forms sampled item by item, mean over severities; for real respondents, sampled the same way from their answers in each band (range over bands). First r: correlation of the first answer with the rest of the form, after removing severity and item means. Lock: % of items whose most common (whole forms) or most likely (log-probabilities) answer equals the severity level (mild 1, moderate 2, severe 3; moderately severe has no single level and is left out), label (L) or impairment (I) framing.

	Main runs		API default	
Model	Generated	Reasoning	Generated	Reasoning
GPT-4o-mini	46	0	—	—
GPT-4o	46	0	—	—
GPT-4.1-nano	48	0	—	—
GPT-4.1-mini	46	0	—	—
GPT-4.1	46	0	—	—
GPT-5-mini	59	0	525	466
GPT-5.2	53	0	—	—
GPT-5.4-mini	53	0	—	—
Haiku 4.5	66	—	—	—
Grok-4.1	46	0	—	—
Llama-3.3	64	0	—	—
Kimi-K2.5	64	—	1,760	—
Mistral-L3	74	0	—	—
Phi-4	71	0	—	—
Gemini-2.5-FL	78	0	—	—
Gemini-2.5-F	63	0	810	749
Gemini-2.5-Pro	121	68	1,368	1,304

Table 16: Tokens per PHQ-9 form, mean. Main runs: reasoning off, or at the minimum thinking budget (128 tokens) for Gemini 2.5 Pro. API default: for the four models whose default reasons. Generated: all tokens produced, including reasoning. Reasoning: as itemised by the API (—, not itemised or not run).

			PHQ-9					GAD-7			PCL-5 roles			Screening	
Model	Effort	Tokens	Calib.	First r	All 9	Div.	AUC	All 7	Div.	AUC	Ratio	AUC	Δtotal	AUC	FPR
GPT-5-mini	minimal*	0	52	0.31	87	1.31	0.77	100	1.14	0.76	0.03	0.66	28.6	0.79	5.0
	low	190	98	−0.08	88	2.15	0.83	99	2.20	0.67	0.01	0.74	−37.1	0.75	10.9
	medium†	466	98	0.03	93	1.49	0.84	99	2.28	0.62	0.00	0.65	−46.4	0.80	8.0
	high	1,149	99	0.01	91	1.77	0.85	100	2.12	0.66	0.08	0.63	−55.2	0.77	7.7
Gemini-2.5-F	off*	0	42	0.35	93	1.75	0.79	100	1.87	0.63	0.04	0.77	14.5	0.75	10.8
	low	735	38	0.46	97	1.56	0.80	100	0.90	0.62	0.10	0.69	16.1	0.78	10.6
	dynamic†	749	34	0.47	94	1.14	0.84	100	1.18	0.64	0.08	0.74	17.0	0.78	9.3
	high	1,139	69	0.60	94	1.26	0.86	100	0.67	0.64	0.07	0.64	14.5	0.74	11.9
Kimi-K2.5	off*	64‡	57	0.54	89	1.44	0.74	100	1.45	0.71	0.13	0.77	12.0	0.80	8.0
	on†	1,760‡	95	0.15	90	1.77	0.77	100	1.51	0.70	0.20	0.77	8.9	0.79	5.5
Gemini-2.5-Pro	min*	68	49	0.34	88	1.83	0.68	100	1.65	0.68	0.37	0.84	14.3	0.75	8.0
	dynamic†	1,304	87	−0.01	99	1.35	0.75	100	1.68	0.59	0.15	0.73	12.4	0.75	13.3

Table 17: Reasoning effort; *main run (reasoning off, or at the minimum for Gemini 2.5 Pro), †API default; off: no thinking; min: a thinking budget of 128 tokens. Tokens: mean reasoning tokens per PHQ-9 form (‡generated tokens; Kimi-K2.5 does not itemise reasoning). First r: as in Table 15. PHQ-9 (totals 15–24) and GAD-7 (12–19) measures as in Tables 4 and 5. PCL-5 roles: "not at all" ratio and separability of the genuine-patient role against the genuinely injured students, and the change in total from the patient to the feigning role, as in Table 13. Screening: model trained on the PHQ-9 responses, as in Table 31.

Model	Effort	PHQ-9	COACH	GAD-7	K6	PHQ-15	OCI-R	ITQ	PCL-5 pp.	EDE-Q
GPT-5-mini	minimal*	0.78 / 6.9	0.72 / 6.8	0.71 / 6.5	0.68 / 6.1	0.89 / 4.3	0.82 / 4.0	0.70 / 3.6	0.92 / —	0.94 / 5.5
	low	0.79 / 2.9	0.75 / 2.7	0.66 / 3.2	0.69 / 4.9	0.89 / 4.7	0.84 / 6.3	0.69 / 3.2	0.93 / —	0.91 / 5.3
	medium†	0.80 / 3.6	0.77 / 3.3	0.62 / 2.8	0.67 / 4.7	0.90 / 4.2	0.86 / 6.0	0.67 / 3.5	0.90 / —	0.92 / 5.4
	high	0.81 / 3.6	0.77 / 3.2	0.62 / 3.1	0.67 / 4.0	0.90 / 2.1	0.89 / 3.3	0.63 / 3.9	0.88 / —	0.91 / 5.5
Gemini-2.5-F	off*	0.79 / 5.4	0.80 / 4.8	0.67 / 4.1	0.64 / 3.7	0.90 / 3.1	0.93 / 2.0	0.64 / 2.1	0.95 / —	0.91 / 4.5
	low	0.80 / 4.9	0.81 / 4.4	0.64 / 8.0	0.66 / 6.8	0.89 / 4.1	0.95 / 3.2	0.69 / 3.8	0.92 / —	0.90 / 3.5
	dynamic†	0.81 / 5.6	0.80 / 5.3	0.65 / 7.2	0.66 / 5.9	0.90 / 3.4	0.95 / 3.9	0.69 / 4.0	0.93 / —	0.88 / 2.9
	high	0.83 / 3.8	0.82 / 3.4	0.65 / 8.2	0.66 / 7.1	0.90 / 2.9	0.96 / 3.4	0.70 / 4.3	0.92 / —	0.90 / 3.3
Kimi-K2.5	off*	0.72 / 7.6	0.78 / 6.8	0.71 / 5.3	0.80 / 4.5	0.93 / 2.3	0.96 / 2.8	0.72 / 2.6	0.94 / —	0.92 / 4.2
	on†	0.75 / 4.4	0.74 / 4.4	0.68 / 4.1	0.73 / 3.6	0.93 / 1.8	0.98 / 3.2	0.84 / 3.2	0.91 / —	0.91 / 5.2
Gemini-2.5-Pro	min*	0.66 / 4.6	0.81 / 4.0	0.72 / 4.5	0.68 / 4.0	0.97 / 2.7	0.99 / 2.1	0.75 / 2.2	0.97 / —	0.90 / 4.2
	dynamic†	0.76 / 4.6	0.82 / 4.2	0.60 / 3.8	0.70 / 5.2	0.94 / 2.0	0.98 / 1.8	0.70 / 2.5	0.96 / —	0.90 / 3.6

Table 18: Reasoning effort on the nine references, protocol of Table 1; efforts marked as in Table 17. Each cell: separability (AUC) / real-to-model diversity ratio, shown where at least five totals carry it. Clinical samples: COACH PHQ-9 and postpartum PCL-5.

		Interventions (AUC)		Published personas			Interviews (rater GPT-4.1)				
Model	Effort	One-line	+5 sheets	TalkDep	CACTUS	SI	All 8	Div.	r real	r sampled	SI agr.
Gemini-2.5-F	off*	0.81	0.62	0.85	0.88	16	100 / 83	0.69 / 5.40	0.83	0.79	100
	low	0.80	0.75	0.85	0.87	9	100 / 74	0.83 / 5.41	0.89	0.89	100
	dynamic†	0.79	0.72	0.89	0.90	7	100 / 77	0.77 / 5.34	0.90	0.89	100
	high	0.83	0.74	0.88	0.91	6	99 / 77	0.48 / 5.94	0.89	0.88	100
Gemini-2.5-Pro	min*	0.64	0.69	0.86	0.88	2	100 / 83	0.71 / 5.50	0.83	0.83	100
	dynamic†	0.64	0.71	0.80	0.88	5	100 / 83	0.27 / 4.35	0.85	0.82	100

Table 19: Reasoning effort in the Gemini 2.5 Flash and Pro experiments; efforts marked as in Table 17. Interventions and published personas as in Tables 20 and 21 (AUC at totals 15–24 against held-out real respondents; SI: % endorsing item 9 with CACTUS personas). Interviews as in Table 25: severity-persona interviews, all eight endorsed and diversity, each against real respondents at the same totals; r: correlation of given and rated answers with a real or a sampled profile; SI agr.: % of real-profile interviews whose rated suicidal ideation matches the profile.

Model	Condition	Calib.	Div.	Real$_=$	N.a.a.	AUC	SI$_{ms/s}$
GPT-4o-mini	One-line	46	2.23	8.09	0.0	0.88	100
	+ "people differ"	42	2.49	7.96	0.1	0.86	100
	+ biography	49	1.95	8.25	0.0	0.86	99
	Total, no label	68	2.28	8.44	0.2	0.82	78
	+ 5 real sheets	63	4.51	8.10	0.5	0.69	86
GPT-4.1	One-line	48	0.66	6.69	0.0	0.74	100
	+ "people differ"	73	1.89	7.34	0.0	0.76	98
	+ biography	52	1.17	7.04	0.0	0.71	100
	Total, no label	100	3.01	7.72	0.2	0.69	88
	+ 5 real sheets	83	3.16	7.85	0.0	0.57	92
Haiku 4.5	One-line	25	1.05	7.32	0.0	0.85	100
	+ "people differ"	48	2.28	8.36	0.1	0.84	99
	+ biography	41	1.34	7.59	0.0	0.81	100
	Total, no label	60	2.43	8.13	0.0	0.79	34
	+ 5 real sheets	88	2.27	7.41	0.0	0.64	81
Llama-3.3	One-line	42	0.63	7.28	0.0	0.73	100
	+ "people differ"	46	2.45	8.01	0.1	0.86	82
	+ biography	49	1.26	7.40	0.0	0.80	86
	Total, no label	47	2.76	7.84	—	—	6
	+ 5 real sheets	76	3.43	7.10	0.0	0.68	75
Gemini-2.5-FL	One-line	33	2.03	6.95	0.1	0.87	100
	+ "people differ"	45	2.28	7.96	0.1	0.87	99
	+ biography	40	2.24	7.83	0.2	0.85	94
	Total, no label	28	2.47	7.37	0.0	—	21
	+ 5 real sheets	82	4.08	7.95	0.8	0.68	70
Gemini-2.5-F	One-line	29	0.98	6.07	0.0	0.81	100
	+ "people differ"	51	1.95	7.57	0.5	0.79	98
	+ biography	30	1.87	7.48	0.1	0.76	100
	Total, no label	99	4.00	7.79	1.3	0.65	70
	+ 5 real sheets	84	3.64	7.82	1.3	0.62	76
Gemini-2.5-Pro	One-line	39	1.75	7.12	0.0	0.64	100
	+ "people differ"	64	3.57	8.18	6.5	0.77	89
	+ biography	47	1.76	7.73	0.1	0.72	98
	Total, no label	100	3.11	7.72	0.4	0.59	70
	+ 5 real sheets	88	4.52	7.98	5.5	0.69	82

Table 20: Interventions (PHQ-9). Calib.: % in the instructed band. Div.: diversity, totals 5–27; Real$_=$: held-out real respondents at the same totals. N.a.a.: % "not at all" on items 1–8, totals 15–24 (real: 9.9). AUC: separability from held-out real respondents (chance 0.44–0.57; —, too few responses at totals 15–24). SI$_{ms/s}$: % endorsing item 9 among moderately severe and severe personas, all totals.

Model	Persona	n	Div.	Real$_=$	N.a.a. 1–8	SI	All 9	AUC	AUC$_{1-8}$
GPT-4o-mini	One-line	160	2.23	8.09	0.0	99.1	99.1	0.88	0.85
	TalkDep	360	3.95	7.94	3.6	43.1	29.2	0.89	0.88
	CACTUS	400	3.07	8.79	1.1	1.0	1.0	0.92	0.88
	CACTUS + form.	400	2.97	8.73	1.3	3.5	2.5	0.92	0.91
GPT-4.1	One-line	160	0.66	6.69	0.0	100.0	100.0	0.74	0.71
	TalkDep	360	2.39	7.39	0.0	72.0	72.0	0.75	0.74
	CACTUS	400	2.74	8.22	0.0	17.2	17.2	0.84	0.82
	CACTUS + form.	400	2.66	8.27	0.0	19.9	19.9	0.86	0.84
Haiku 4.5	One-line	160	1.05	7.32	0.0	94.3	94.3	0.85	0.83
	TalkDep	360	2.63	7.27	0.0	51.8	51.8	0.88	0.89
	CACTUS	400	3.21	7.96	0.0	8.0	8.0	0.86	0.81
	CACTUS + form.	400	3.18	8.22	0.0	8.0	8.0	0.89	0.83
Llama-3.3	One-line	160	0.63	7.28	0.0	83.6	83.6	0.73	0.80
	TalkDep	360	2.31	7.27	0.0	32.0	32.0	0.86	0.84
	CACTUS	400	3.08	8.19	0.2	0.0	0.0	0.90	0.82
	CACTUS + form.	400	3.00	8.19	0.3	0.0	0.0	0.92	0.86
Gemini-2.5-FL	One-line	160	2.03	6.95	0.1	88.2	87.4	0.87	0.85
	TalkDep	360	3.49	7.64	0.8	48.3	45.5	0.88	0.91
	CACTUS	400	3.36	8.37	1.5	2.8	1.9	0.92	0.88
	CACTUS + form.	400	3.35	8.31	2.0	0.7	0.7	0.93	0.87
Gemini-2.5-F	One-line	160	0.98	6.07	0.0	93.1	93.1	0.81	0.77
	TalkDep	360	3.41	7.26	0.3	44.5	44.5	0.85	0.86
	CACTUS	400	3.43	8.19	2.7	15.8	12.9	0.88	0.83
	CACTUS + form.	400	3.25	8.18	2.1	18.3	17.3	0.91	0.87
Gemini-2.5-Pro	One-line	160	1.75	7.12	0.0	90.5	90.5	0.64	0.62
	TalkDep	360	3.02	7.29	0.0	49.3	49.3	0.86	0.87
	CACTUS	400	3.18	8.19	1.7	1.8	0.9	0.88	0.81
	CACTUS + form.	400	3.04	8.32	1.2	0.7	0.7	0.90	0.84
Real (held-out)		707	—	—	9.9	42.3	18.5	—	—

Table 21: Published simulator personas (PHQ-9), intervention protocol. One-line: the severity persona, from the intervention runs. CACTUS + form.: the CACTUS intake form with the automatic thought and cognitive distortions of the client's case formulation. n: responses; for Real, held-out respondents at totals 15–24. Div.: diversity, totals 5–27; Real$_=$: held-out real respondents at the same totals. At totals 15–24: N.a.a. 1–8, % "not at all" on items 1–8; SI, % endorsing item 9; All 9, % endorsing every item; AUC, separability from held-out real respondents (chance 0.44–0.55); AUC$_{1-8}$, the same without item 9. The reference is the held-out half of NHANES, so one-line values differ slightly from Table 4. TalkDep personas are each answered repeatedly, so their diversity uses pairs of different personas and a persona's responses stay in one cross-validation fold.

					Breadth		Several days		Key	Other	
Persona	BDI-II	PHQ-9	s.d.	Distinct	Model	Real	Model	Real	Model	Model	Real
Noah	5	4.7	0.7	23	4.7	3.6	100	78	91	33	36
Maya	6	3.3	0.7	20	3.3	2.6	100	83	97	7	27
Priya	7	4.7	0.8	23	4.7	3.6	100	78	100	30	36
Ethan	12	8.3	0.9	27	6.3	5.4	67	61	100	54	55
Gabriel	13	10.7	1.1	37	7.8	6.2	62	50	100	82	68
Alex	15	9.4	0.9	27	7.6	5.7	77	56	100	77	60
James	22	14.6	0.7	20	8.0	7.0	29	30	100	83	74
Laura	23	16.4	1.1	37	8.1	7.4	26	25	100	82	76
Linda	28	16.4	1.2	40	8.0	7.5	24	25	100	81	76
Elena	35	22.4	1.3	43	9.0	8.4	8	8	100	100	91
Marco	38	23.5	0.9	30	9.0	8.5	8	6	100	100	93
Maria	40	16.8	1.4	47	8.5	7.5	30	23	100	91	74

Table 22: TalkDep personas, median over 17 models. BDI-II: published score. PHQ-9: mean total over 30 runs, with its s.d. Distinct: % distinct answer patterns. Breadth: symptoms endorsed. Several days: % of endorsed answers at "several days". Key, Other: % endorsing the items matching the persona's key symptoms, and the rest. Real: real respondents at the same totals.

	PHQ-9				GAD-7			
Model	One-line	TalkDep	CACTUS	+ form.	One-line	TalkDep	CACTUS	+ form.
GPT-4o-mini	0.83 / 4.3	0.81 / 2.0	0.93 / 2.9	0.94 / 3.0	0.70 / 4.8	0.63 / 2.3	0.66 / 3.1	0.70 / 3.1
GPT-4o	0.77 / 3.4	0.72 / 2.3	0.89 / 2.4	0.89 / 2.6	0.76 / 3.8	0.54 / 2.1	0.72 / 2.8	0.74 / 2.9
GPT-4.1-nano	0.83 / 3.2	0.76 / 1.6	0.88 / 2.3	0.90 / 2.3	0.72 / 4.6	0.58 / 1.9	0.62 / 2.5	0.58 / 2.2
GPT-4.1-mini	0.77 / 4.3	0.73 / 2.1	0.88 / 2.7	0.90 / 3.2	0.72 / 4.4	0.62 / 2.3	0.61 / 3.4	0.64 / 3.2
GPT-4.1	0.67 / 7.7	0.71 / 3.1	0.89 / 3.0	0.88 / 3.1	0.71 / 5.0	0.49 / 2.6	0.64 / 3.4	0.68 / 4.0
GPT-5-mini	0.75 / 6.0	0.76 / 2.5	0.89 / 2.9	0.90 / 3.3	0.69 / 5.7	0.55 / 3.2	0.63 / 4.2	0.67 / 4.1
GPT-5.2	0.75 / 9.7	0.71 / 3.0	0.89 / 2.8	0.89 / 2.8	0.71 / 9.1	0.57 / 3.6	0.66 / 4.2	0.67 / 4.5
GPT-5.4-mini	0.75 / 6.3	0.71 / 2.7	0.92 / 2.6	0.92 / 2.8	0.67 / 5.9	0.52 / 3.7	0.69 / 4.8	0.71 / 4.0
Haiku 4.5	0.80 / 7.3	0.73 / 2.8	0.91 / 2.5	0.92 / 2.6	0.70 / 4.8	0.64 / 3.4	0.58 / 3.3	0.60 / 3.3
Grok-4.1	0.76 / 4.9	0.62 / 2.0	0.91 / 2.1	0.92 / 2.3	0.69 / 5.2	0.70 / 2.4	0.75 / 2.6	0.76 / 2.3
Llama-3.3	0.80 / 6.2	0.78 / 3.2	0.92 / 2.7	0.93 / 2.7	0.66 / 6.3	0.66 / 3.5	0.67 / 3.3	0.64 / 3.3
Kimi-K2.5	0.72 / 5.8	0.76 / 2.7	0.89 / 2.5	0.90 / 2.5	0.70 / 4.6	0.56 / 3.0	0.70 / 3.0	0.72 / 3.1
Mistral-L3	0.72 / 5.7	0.72 / 3.1	0.89 / 2.5	0.89 / 2.6	0.66 / 5.7	0.71 / 3.5	0.64 / 3.7	0.65 / 3.8
Phi-4	0.81 / 4.2	0.75 / 2.3	0.93 / 2.3	0.94 / 2.5	0.65 / 4.9	0.72 / 2.5	0.73 / 2.8	0.68 / 2.8
Gemini-2.5-FL	0.84 / 3.7	0.81 / 2.2	0.93 / 2.5	0.95 / 2.5	0.70 / 4.0	0.69 / 2.9	0.65 / 3.4	0.68 / 3.2
Gemini-2.5-F	0.79 / 5.0	0.74 / 2.1	0.92 / 2.4	0.92 / 2.5	0.66 / 3.5	0.65 / 2.8	0.65 / 2.4	0.66 / 2.3
Gemini-2.5-Pro	0.67 / 4.7	0.78 / 2.4	0.93 / 2.6	0.94 / 2.8	0.72 / 4.4	0.66 / 2.5	0.68 / 3.1	0.67 / 3.1

Table 23: Published simulator personas on every model and on both questionnaires. Each cell: separability (AUC) from real respondents with the same totals (NHANES for the PHQ-9, NHIS for the GAD-7; totals $\geq$5) / real-to-model diversity ratio. One-line: the severity persona of the main runs. + form.: CACTUS with the automatic thought and cognitive distortions of the client's case formulation. All totals of at least 5 are used, so one-line values differ slightly from Tables 4, 5 and 21 (totals 15–24). TalkDep personas are each answered repeatedly, so their diversity uses pairs of different personas and a persona's responses stay in one cross-validation fold.

Family	Model	Diversity [2.5th, 97.5th]	Calib. (%)
GPT-4.1	GPT-4.1-nano	2.32 [2.23, 2.44]	46.8
GPT-4.1	GPT-4.1-mini	1.86 [1.76, 1.94]	49.1
GPT-4.1	GPT-4.1	0.88 [0.77, 0.94]	65.9
GPT-4o	GPT-4o-mini	1.95 [1.87, 2.06]	41.8
GPT-4o	GPT-4o	2.43 [2.34, 2.51]	58.6
GPT-5	GPT-5-mini	1.31 [1.11, 1.40]	52.3
GPT-5	GPT-5.2	0.83 [0.76, 0.91]	50.7
Gemini-2.5	Gemini-2.5-FL	2.06 [1.97, 2.15]	48.2
Gemini-2.5	Gemini-2.5-F	1.75 [1.56, 1.82]	41.6
Gemini-2.5	Gemini-2.5-Pro	1.83 [1.76, 2.05]	48.9
Real		7.35	—

Table 24: Diversity by model tier within families (PHQ-9, totals 15–24), with the 2.5th–97.5th percentiles over 80% subsamples drawn without replacement.

Patient	Rater	n	Breadth (/8)	All 8 (%)	Real$_=$	Not at all (%)	Div.	Real$_=$	SI (%)	SI agr.	r_{item}	r_{total}
NHANES, total $\geq$5	—	10760	5.25	10.7	—	34.3	6.21	—	14.3	—	—	—
severity, Haiku 4.5	GPT-4.1	80	7.99	98.8	81.8	0.2	1.20	5.79	82.5	—	—	—
severity, Gemini-2.5-F	GPT-4.1	80	8.00	100.0	82.8	0.0	0.69	5.40	97.5	—	—	—
severity, Gemini-2.5-F	Haiku 4.5	80	8.00	100.0	67.4	0.0	0.60	6.11	97.5	—	—	—
severity, Gemini-2.5-FL	GPT-4.1	80	8.00	100.0	72.2	0.0	0.79	5.65	75.0	—	—	—
severity, Gemini-2.5-FL	Haiku 4.5	80	8.00	100.0	58.0	0.0	1.01	6.25	75.0	—	—	—
severity, Gemini-2.5-Pro	GPT-4.1	80	8.00	100.0	82.9	0.0	0.71	5.50	77.5	—	—	—
severity, Gemini-2.5-Pro	Haiku 4.5	80	8.00	100.0	62.1	0.0	0.36	5.33	77.5	—	—	—
severity, GPT-4.1	Haiku 4.5	80	7.97	97.5	53.4	0.3	0.57	5.47	88.8	—	—	—
severity, GPT-4o-mini	GPT-4.1	80	8.00	100.0	67.2	0.0	0.97	5.90	97.5	—	—	—
severity, GPT-4o-mini	Haiku 4.5	80	8.00	100.0	48.8	0.0	0.85	6.61	96.2	—	—	—
severity, Llama-3.3	GPT-4.1	71	8.00	100.0	84.8	0.0	0.48	4.65	—	—	—	—
severity, Llama-3.3	Haiku 4.5	71	8.00	100.0	56.9	0.0	0.82	6.39	—	—	—	—
real profile, Gemini-2.5-F	GPT-4.1	80	6.67	35.0	38.8	16.6	4.91	6.83	31.2	100	0.83	0.90
real profile, Gemini-2.5-F	Haiku 4.5	80	6.70	35.0	33.7	16.2	4.81	7.09	31.2	100	0.82	0.90
real profile, Gemini-2.5-FL	GPT-4.1	80	6.75	37.5	49.7	15.6	4.03	6.63	31.2	100	0.75	0.83
real profile, Gemini-2.5-FL	Haiku 4.5	80	6.79	40.0	39.0	15.2	4.25	7.22	31.2	100	0.73	0.81
real profile, Gemini-2.5-Pro	GPT-4.1	80	6.64	35.0	44.8	17.0	4.64	6.41	31.2	100	0.83	0.89
real profile, Gemini-2.5-Pro	Haiku 4.5	80	6.61	35.0	37.7	17.3	4.53	6.81	31.2	100	0.82	0.88
real profile, GPT-4o-mini	GPT-4.1	80	6.81	40.0	41.7	14.8	3.59	6.55	31.2	100	0.78	0.89
real profile, GPT-4o-mini	Haiku 4.5	80	6.79	40.0	31.5	15.2	3.30	7.25	31.2	100	0.74	0.86
real profile, Llama-3.3	GPT-4.1	74	6.77	36.5	50.2	15.4	4.57	6.77	—	—	0.72	0.83
real profile, Llama-3.3	Haiku 4.5	74	6.74	35.1	37.7	15.7	4.46	7.34	—	—	0.70	0.83
sampled profile, Gemini-2.5-F	GPT-4.1	80	6.81	42.5	43.7	14.8	4.86	7.01	36.2	100	0.79	0.90
sampled profile, Gemini-2.5-F	Haiku 4.5	80	6.83	42.5	37.3	14.7	4.30	7.04	36.2	100	0.79	0.90
sampled profile, Gemini-2.5-FL	GPT-4.1	80	6.89	46.2	50.7	13.9	3.47	6.95	36.2	100	0.73	0.85
sampled profile, Gemini-2.5-FL	Haiku 4.5	80	6.88	45.0	38.9	14.1	3.41	6.98	36.2	100	0.71	0.81
sampled profile, Gemini-2.5-Pro	GPT-4.1	80	6.74	42.5	45.8	15.8	4.13	6.09	36.2	100	0.83	0.89
sampled profile, Gemini-2.5-Pro	Haiku 4.5	80	6.76	42.5	38.6	15.5	4.42	6.74	36.2	100	0.82	0.90
sampled profile, GPT-4o-mini	GPT-4.1	80	6.86	43.8	46.1	14.2	3.93	7.16	36.2	100	0.75	0.89
sampled profile, GPT-4o-mini	Haiku 4.5	80	6.88	46.2	35.7	14.1	3.26	7.12	36.2	100	0.73	0.87
sampled profile, Llama-3.3	GPT-4.1	72	6.78	41.7	53.1	15.3	3.81	6.58	—	—	0.71	0.85
sampled profile, Llama-3.3	Haiku 4.5	72	6.76	41.7	40.3	15.5	3.51	6.90	—	—	0.70	0.85

Table 25: Interviews, PHQ-9 items 1–8 rated from the transcript. Severity: the patient is told a severity; real profile: given a real NHANES profile; sampled profile: given a GRM-sampled profile. Real$_=$: real NHANES respondents at the same item 1–8 totals. SI agr.: % of interviews whose rated suicidal ideation (present or absent) matches the given profile. r: correlation of given and rated profile. SI is not reported for Llama-3.3 (Appendix C.1). The two raters agree within one point on 95–100% of items.

Given	Rater	Items	Exact	Within 1	κ_w	Present→present	Absent→absent	Exact if present	1 rated higher	Shift
Real profile	GPT-4.1	3,152	56.6	92.4	0.71	100.0	84.8	50.1	80.7	+0.31
Real profile	Haiku 4.5	3,152	54.2	93.5	0.71	100.0	84.9	47.1	76.6	+0.20
Sampled profile	GPT-4.1	3,136	53.4	90.4	0.68	100.0	83.2	46.9	84.6	+0.36
Sampled profile	Haiku 4.5	3,136	52.2	92.5	0.68	100.0	82.8	45.6	80.5	+0.25

Table 26: Given against rated answers, PHQ-9 items 1–8, pooled over the patient models (a model never rates its own transcripts). Items: given answers compared. Exact, Within 1: % of items rated at the given level, or within one level. κ_w: quadratically weighted kappa. Present→present: % of symptoms given as present (above 0) that are rated present; Absent→absent: % of symptoms given at 0 that are rated 0. Exact if present: % of present symptoms rated at the given level. 1 rated higher: % of symptoms given at "several days" rated higher. Shift: mean rated minus given level.

	Real-to-model diversity		All 8 endorsed (%)		AUC against real respondents		
Patient model	Questionnaire	Interview	Severity	Real profile	Severity	Real profile	Sampled profile
Haiku 4.5	7.7	—	—	—	0.66	—	—
Gemini-2.5-F	4.2	3.7	100	57	0.72	0.66	0.72
Gemini-2.5-FL	3.6	3.0	100	46	0.73	0.70	0.79
Gemini-2.5-Pro	4.0	4.6	100	50	0.68	0.65	0.62
GPT-4.1	8.3	—	—	—	0.78	—	—
GPT-4o-mini	4.0	2.9	100	59	0.73	0.69	0.72
Llama-3.3	5.1	3.6	100	47	0.69	0.73	0.76

Table 27: Interviews against real profiles passed through the same interviewer and rater (PHQ-9 items 1–8). Questionnaire: real-to-model diversity ratio at totals 15–24, as in Table 4. Interview: diversity of real-profile interviews over severity-persona interviews, both computed within the same bins of three total points (bins with at least three interviews of each kind). All 8: % of interviews rated as endorsing all eight items, mean over the same bins. AUC: separability of the rated profiles from real NHANES respondents with the same item 1–8 totals. —: no real-profile interviews for this model.

			All 8 (%)		Diversity				Agent against the label			ρ with BDI-II	
Patient model	n	Breadth (/8)	TalkDep	Real$_=$	TalkDep	Real$_=$	Ratio	AUC	Exact	Within 1	Above	Interviews	Personas
GPT-4o-mini	84	6.82	83	58	2.14	6.72	1.9	0.67	37	87	62	0.90	0.93
GPT-4.1	84	6.33	95	45	2.09	6.72	—	0.71	62	94	38	0.94	0.96
Haiku 4.5	84	6.63	95	80	2.55	5.98	—	0.73	40	82	60	0.95	0.96
Llama-3.3	72	7.25	100	72	1.59	6.49	2.2	0.63	22	72	78	0.90	0.94
Gemini-2.5-FL	84	6.54	86	50	2.40	6.67	1.6	0.66	51	98	46	0.87	0.89
Gemini-2.5-F	84	6.32	81	52	2.31	6.28	1.7	0.70	60	92	35	0.87	0.91
Gemini-2.5-Pro	84	6.43	81	60	2.14	6.08	2.1	0.64	57	89	42	0.91	0.91

Table 28: Interviews with the 12 TalkDep personas, seven per persona, PHQ-9 items 1–8 rated by the rater used in the paper. All 8: % rated as endorsing all eight items, personas labelled moderate or above. Real$_=$: real NHANES respondents at the same item 1–8 totals. Diversity and Ratio use pairs of interviews with different personas; Ratio: diversity of real-profile interviews over TalkDep interviews within the same bins of three total points, as in Table 27 (—: no real-profile interviews). AUC: separability from real respondents with the same item 1–8 totals, the interviews of one persona in one cross-validation fold. Agent: rated PHQ-9 band against the band named like the persona's severity label (minimal: none 0–4; mild 5–9; moderate 10–14; moderate to severe: 15–19; severe 20–27), % exact, within one band and above. ρ: Spearman correlation of the rated total with the persona's BDI-II score, over interviews and over the persona means.

	Against the given profile (50 interviews)							Severity persona
Rater	Exact	Within 1	κ_w	Absent→absent	Exact if present	1 rated higher	Shift	All 8 (%)
R1	55.2	92.8	0.71	89.9	46.7	68	+0.05	100
R2	57.0	92.0	0.70	89.9	48.9	70	+0.13	100
R3	58.2	89.0	0.68	89.9	50.5	75	+0.22	100
R4	55.5	92.8	0.70	87.3	47.7	72	+0.15	100
R5	54.0	87.5	0.65	87.3	45.8	89	+0.34	100
All five	56.0	90.8	0.69	88.9	47.9	75 [67, 81]	+0.18	100
Median of five	57.8	92.0	0.70	89.9	49.8	73	+0.17	—
Model rater	53.8	89.5	0.67	87.3	45.5	91 [85, 97]	+0.39	100

Table 29: Human ratings of 64 interview transcripts, PHQ-9 items 1–8 (Appendix C). Columns 2–8: the 40 real-profile and 10 sampled-profile interviews against the profile given to the patient model, as in Table 26; 1 rated higher: % of symptoms given as "several days" rated higher, with 95% bootstrap intervals over transcripts. All 8: % of the 14 severity-persona interviews rated as endorsing all eight items. Model rater: the rater used in the paper, on the same transcripts. Agreement between the human raters, all pairs: exact 62–76%, κ_w 0.79–0.87; each human rater against the model rater: κ_w 0.78–0.89, the model rater higher by 0.07–0.38 levels on average.

Source	n	Breadth	All 9	N.a.a.	SI	Div.	AUC
Real, 2013–2023 (held out)	5,779	5.43	4.1	39.6	14.0	7.30	—
GRM fit on 2005–2012	5,779	5.49	4.9	39.1	15.6	7.23	0.52
Real, 2005–2012	4,981	5.38	4.2	40.2	14.9	7.01	0.53

Table 30: Profiles sampled from a GRM fit to NHANES 2005–2012 (Appendix C), selected to match the total distribution of the held-out 2013–2023 respondents (totals $\geq$5); 2005–2012 respondents are reweighted to it. Div.: diversity. AUC: separability from the held-out respondents.

Trained on	n	SI	AUC	Brier	Slope	Sens.	FPR	Recal. Brier
Real (NHANES 2005–2012)	19,685	3.2	0.82	0.096	0.83	25.6	3.1	0.096
GRM-sampled profiles	20,000	4.3	0.79	0.100	0.81	25.6	3.7	0.099
GPT-4o-mini	450	64.7	0.80	0.132	0.18	44.5	8.2	0.098
GPT-4o	437	59.0	0.79	0.129	0.23	42.5	8.4	0.100
GPT-4.1-nano	436	76.8	0.72	0.240	0.21	63.4	28.1	0.106
GPT-4.1-mini	436	50.9	0.78	0.123	0.22	30.3	5.4	0.101
GPT-4.1	437	62.2	0.75	0.152	0.19	48.1	12.3	0.103
GPT-5-mini	436	48.6	0.79	0.118	0.23	31.4	5.0	0.099
GPT-5.2	450	66.4	0.78	0.144	0.18	48.3	10.8	0.100
GPT-5.4-mini	450	50.0	0.81	0.114	0.27	31.7	4.6	0.097
Haiku 4.5	450	54.0	0.77	0.128	0.17	34.2	5.9	0.101
Grok-4.1	450	35.6	0.77	0.122	0.19	22.1	3.4	0.102
Llama-3.3	450	35.1	0.76	0.130	0.18	36.9	7.1	0.102
Kimi-K2.5	450	43.6	0.80	0.126	0.23	42.7	8.0	0.098
Mistral-L3	450	54.9	0.76	0.135	0.17	38.0	8.2	0.102
Phi-4	450	58.7	0.76	0.121	0.26	26.7	4.6	0.103
Gemini-2.5-FL	450	65.6	0.80	0.133	0.26	49.9	10.5	0.098
Gemini-2.5-F	450	62.2	0.75	0.147	0.19	42.7	10.8	0.104
Gemini-2.5-Pro	450	55.8	0.75	0.135	0.18	37.9	8.0	0.103
All models pooled	7,582	55.5	0.79	0.127	0.25	40.4	8.1	0.100

Table 31: Predicting whether PHQ-9 item 9 is endorsed from items 1–8. Test set: NHANES 2013–2023 respondents with items 1–8 total $\geq$5 (n=5,747). n, SI: size of the training set and % endorsing item 9 in it. GRM-sampled: profiles from the GRM fit to NHANES 2005–2012. Slope: calibration slope (1 when calibrated). Sens., FPR: sensitivity and false-positive rate at $p \geq 0.5$ (%). Recal. Brier: after logistic recalibration on 1,000 held-out 2005–2012 respondents with items 1–8 total $\geq$5.

	SI in training (%)		AUC				AUC difference from real		Brier	
Simulated patients	Sim.	Real_t	Sim.	Real_r	Real_t	$\text{Real}_{t,9}$	Real_t	$\text{Real}_{t,9}$	Sim. recal.	Real_t
GPT-4o-mini	65	28	0.803	0.764	0.808	0.775	−0.005 [−0.030, +0.023]	+0.028 [+0.008, +0.045]	0.098	0.098
GPT-4o	59	30	0.786	0.769	0.803	0.780	−0.017 [−0.059, +0.017]	+0.006 [−0.032, +0.031]	0.100	0.099
GPT-4.1-nano	77	29	0.721	0.767	0.801	0.786	−0.080 [−0.141, −0.027]	−0.064 [−0.114, −0.025]	0.106	0.099
GPT-4.1-mini	51	33	0.780	0.772	0.805	0.777	−0.025 [−0.073, +0.013]	+0.002 [−0.043, +0.029]	0.101	0.098
GPT-4.1	62	29	0.755	0.777	0.801	0.779	−0.046 [−0.086, −0.008]	−0.024 [−0.059, +0.004]	0.103	0.099
GPT-5-mini	49	31	0.793	0.769	0.808	0.776	−0.015 [−0.057, +0.017]	+0.018 [−0.024, +0.042]	0.099	0.098
GPT-5.2	66	33	0.785	0.775	0.803	0.775	−0.018 [−0.044, +0.010]	+0.010 [−0.007, +0.026]	0.100	0.099
GPT-5.4-mini	50	34	0.812	0.773	0.797	0.778	+0.015 [−0.028, +0.049]	+0.034 [−0.004, +0.054]	0.097	0.100
Haiku 4.5	54	32	0.775	0.778	0.804	0.774	−0.029 [−0.060, +0.003]	+0.001 [−0.020, +0.019]	0.101	0.099
Grok-4.1	36	30	0.767	0.761	0.802	0.778	−0.035 [−0.085, +0.002]	−0.011 [−0.054, +0.018]	0.102	0.099
Llama-3.3	35	28	0.764	0.772	0.801	0.775	−0.037 [−0.076, +0.002]	−0.010 [−0.042, +0.017]	0.102	0.098
Kimi-K2.5	44	27	0.801	0.768	0.803	0.778	−0.002 [−0.037, +0.025]	+0.024 [−0.007, +0.038]	0.098	0.098
Mistral-L3	55	31	0.757	0.767	0.809	0.777	−0.052 [−0.090, −0.015]	−0.020 [−0.055, +0.005]	0.102	0.098
Phi-4	59	41	0.764	0.764	0.798	0.790	−0.034 [−0.099, +0.021]	−0.026 [−0.081, +0.012]	0.103	0.100
Gemini-2.5-FL	66	30	0.801	0.770	0.804	0.783	−0.003 [−0.049, +0.028]	+0.018 [−0.020, +0.039]	0.098	0.098
Gemini-2.5-F	62	34	0.748	0.770	0.803	0.776	−0.056 [−0.111, −0.019]	−0.028 [−0.075, +0.001]	0.104	0.099
Gemini-2.5-Pro	56	32	0.755	0.772	0.809	0.774	−0.054 [−0.089, −0.017]	−0.020 [−0.051, −0.000]	0.103	0.098
All models pooled	55	31	0.792	0.818	0.818	0.784	−0.027 [−0.040, −0.015]	+0.008 [−0.003, +0.018]	0.100	0.097

Table 32: Simulated training sets against real training sets of the same size drawn from NHANES 2005–2012: at random (Real_r), with the same items 1–8 totals (Real_t), and with the same totals and the same item-9 answers ($\text{Real}_{t,9}$); control values are means over 50 draws. Differences: simulated minus real, with 95% paired bootstrap intervals (1,000 resamples of the simulated training set and the test set, fresh real draws each time). Sim. recal.: simulated-trained model after recalibration on held-out real respondents. Test set as in Table 31.

Items 1–8	n	Observed	Real	GRM	Sim.
0–4	15,282	0.5	1.1	1.1	0.0
5–9	3,704	6.3	4.2	4.8	0.4
10–14	1,319	17.7	14.2	17.0	13.7
15–19	550	38.5	39.7	46.1	73.3
20–24	174	57.5	68.6	75.2	99.1

Table 33: Observed and predicted rate (%) of endorsing item 9 by total on items 1–8, all NHANES 2013–2023 respondents, for models trained on real respondents, GRM-sampled profiles and simulated patients (pooled over models).

Real n	Added	Draws	Base AUC	ΔAUC	ΔBrier ($\times 10^{-3}$)	ΔPred.
50	+1×	82	0.735	+0.040 [+0.028, +0.053]	+0.1 [−4.0, +4.3]	+2.2
50	+2×	82	0.735	+0.040 [+0.027, +0.053]	+2.9 [−1.3, +7.1]	+1.7
50	+4×	82	0.735	+0.046 [+0.033, +0.059]	+2.0 [−2.8, +6.8]	+0.5
100	+1×	97	0.743	+0.039 [+0.029, +0.049]	+4.9 [+1.9, +8.0]	+3.3
100	+2×	97	0.743	+0.038 [+0.029, +0.048]	+6.5 [+3.5, +9.4]	+2.2
100	+4×	97	0.743	+0.043 [+0.032, +0.053]	+7.8 [+5.0, +10.7]	+1.8
200	+1×	99	0.747	+0.040 [+0.034, +0.047]	+4.0 [+1.8, +6.2]	+3.2
200	+2×	99	0.747	+0.040 [+0.033, +0.048]	+6.1 [+3.7, +8.6]	+2.5
200	+4×	99	0.747	+0.042 [+0.035, +0.050]	+8.0 [+5.5, +10.4]	+1.9
500	+1×	100	0.777	+0.020 [+0.016, +0.025]	+8.5 [+7.1, +10.0]	+2.1
500	+2×	100	0.777	+0.019 [+0.014, +0.024]	+12.3 [+10.8, +13.9]	+1.6
500	+4×	100	0.777	+0.017 [+0.012, +0.022]	+15.4 [+13.8, +16.9]	+1.2

Table 34: Adding simulated patients (pooled over models) to a small random real training set from NHANES 2005–2012. Mean change over draws with its 95% interval; draws without both classes are skipped. ΔPred.: change in mean predicted risk minus the observed rate (points). Test set as in Table 31.

	Told a severity				Given a real profile						Sampled profile	
Patient model	Exact	Within 1	Above	Absent	Exact	Within 1	Above	Absent	Abs. det.	Agree	Exact	Within 1
GPT-4o-mini	25	56	75	0.0	55	96	40	14.8	79	96	56	98
GPT-4.1	26	72	74	0.3	—	—	—	—	—	—	—	—
Haiku 4.5	25	55	75	0.2	—	—	—	—	—	—	—	—
Llama-3.3	20	44	80	0.0	45	91	51	15.4	82	97	49	88
Gemini-2.5-FL	25	56	75	0.0	42	92	51	15.6	83	97	42	91
Gemini-2.5-F	26	51	74	0.0	61	98	34	16.6	88	98	56	96
Gemini-2.5-Pro	25	59	75	0.0	59	94	35	17.0	91	98	64	94
64 transcripts, model rater	33	50	67	0	48	98	48	16	85	97	54	92
64 transcripts, human raters	35 (22–56)	69 (62–88)	60 (44–67)	0 (0–0)	56 (52–68)	97 (92–100)	26 (10–40)	17 (16–17)	87 (85–89)	98 (97–98)	62 (54–71)	98 (92–100)

Table 35: The interviewer and model rater as an assessment agent. Exact, Within 1: % of interviews whose rated PHQ-9 total falls in the patient's severity band (none 0–4, mild 5–9, moderate 10–14, moderately severe 15–19, severe 20–27), or within one band; the patient's severity is the instructed one for patients told a severity and the band of the given profile otherwise. Above: % rated in a higher band. Absent: % of symptoms 1–8 rated absent. Abs. det.: % of symptoms absent from the given profile that are rated absent. Agree: % of symptoms 1–8 whose presence or absence matches the given profile. Bottom rows: the 64 transcripts rated by people (14, 40 and 10 per condition), scored the same way, with Exact, Within 1 and Above averaged with equal weight over the patients' severities, as in the full design; human raters, mean (range) over the five raters. Table 28 gives the TalkDep personas.

Group	Similarity	Contrast
NHANES, random halves	0.98	+0.10
Real, at each model's totals	0.53–0.77	+0.09 to +0.12
Models	0.13–0.61	−0.01 to +0.18
Models' item distributions, no covariation	−0.02 to 0.35	−0.02 to +0.07
Models + noise	0.01–0.12	+0.00 to +0.01
GRM-sampled profiles	0.40	+0.00

Table 36: Item covariance at a fixed PHQ-9 total (totals 5–24). Similarity: correlation of a group's 36 residual correlations with those of all NHANES respondents. Contrast: mean residual correlation within the somatic and within the non-somatic items minus that between them. Ranges over the 17 models. Real respondents drawn at each model's totals, and responses with the model's item distributions at each total and no further covariation, match its sample size (100 draws); GRM-sampled profiles: 20,000. Table 37 gives each model and the interviews.

		Group		Real$_=$		No covariation		Model + noise	
Group	n	Contrast	Similarity	Contrast	Similarity	Contrast	Similarity	Contrast	Similarity
NHANES, random halves	5,352	+0.10	0.98						
GRM-sampled profiles	20,000	+0.00	0.40						
GPT-4o-mini	436	+0.05 [+0.01, +0.09]	0.34	+0.11	0.74 [0.63]	+0.01 [+0.05]	0.21 [0.42]	+0.01	0.09
GPT-4o	360	+0.11 [+0.06, +0.16]	0.61	+0.11	0.71 [0.59]	+0.01 [+0.04]	0.18 [0.41]	+0.01	0.12
GPT-4.1-nano	350	+0.06 [+0.01, +0.10]	0.13	+0.12	0.70 [0.59]	+0.00 [+0.04]	−0.02 [0.20]	+0.01	0.10
GPT-4.1-mini	348	+0.02 [−0.01, +0.06]	0.14	+0.10	0.68 [0.55]	+0.03 [+0.07]	0.20 [0.38]	+0.00	0.04
GPT-4.1	361	+0.03 [−0.03, +0.07]	0.44	+0.11	0.74 [0.62]	−0.02 [+0.02]	0.29 [0.39]	+0.01	0.10
GPT-5-mini	374	+0.09 [+0.04, +0.14]	0.53	+0.10	0.74 [0.62]	+0.00 [+0.05]	0.13 [0.41]	+0.01	0.10
GPT-5.2	394	+0.18 [+0.14, +0.24]	0.46	+0.11	0.69 [0.59]	+0.07 [+0.13]	0.24 [0.41]	+0.01	0.02
GPT-5.4-mini	365	+0.10 [+0.04, +0.15]	0.48	+0.10	0.70 [0.57]	+0.03 [+0.08]	0.24 [0.42]	+0.00	0.05
Haiku 4.5	396	+0.06 [+0.02, +0.12]	0.32	+0.11	0.70 [0.59]	+0.04 [+0.08]	0.21 [0.39]	+0.01	0.04
Grok-4.1	381	−0.01 [−0.04, +0.03]	0.20	+0.11	0.74 [0.64]	+0.01 [+0.05]	0.20 [0.40]	+0.00	0.03
Llama-3.3	363	+0.03 [−0.01, +0.08]	0.25	+0.12	0.77 [0.67]	+0.03 [+0.09]	0.24 [0.39]	+0.00	0.12
Kimi-K2.5	376	+0.10 [+0.05, +0.15]	0.35	+0.12	0.76 [0.65]	+0.03 [+0.07]	0.10 [0.31]	+0.00	0.06
Mistral-L3	366	+0.09 [+0.05, +0.13]	0.54	+0.11	0.72 [0.58]	+0.03 [+0.07]	0.35 [0.49]	+0.01	0.09
Phi-4	351	+0.04 [−0.02, +0.08]	0.22	+0.09	0.53 [0.40]	+0.01 [+0.05]	−0.02 [0.18]	+0.00	0.08
Gemini-2.5-FL	381	+0.10 [+0.06, +0.16]	0.40	+0.11	0.70 [0.59]	+0.00 [+0.05]	0.05 [0.25]	+0.01	0.04
Gemini-2.5-F	349	+0.05 [+0.01, +0.09]	0.38	+0.10	0.66 [0.52]	+0.02 [+0.06]	0.12 [0.33]	+0.00	0.01
Gemini-2.5-Pro	373	+0.07 [+0.03, +0.11]	0.53	+0.11	0.74 [0.65]	+0.01 [+0.05]	0.31 [0.47]	+0.01	0.07
Real profiles given	79	+0.04	0.34	+0.11	0.51 [0.26]				
Interviews, real profile	381	+0.14 [+0.06, +0.20]	0.58	+0.11	0.51 [0.26]				
Sampled profiles given	76	−0.01	0.01	+0.12	0.53 [0.24]				
Interviews, sampled profile	370	+0.04 [−0.07, +0.14]	0.26	+0.12	0.53 [0.24]				

Table 37: Item covariance at a fixed PHQ-9 total (totals 5–24). Residuals are item scores minus the group's mean item scores at the same total. Contrast: mean residual correlation within the somatic (sleep, fatigue, appetite, concentration, psychomotor) and the non-somatic items (interest, mood, worthlessness, suicidal thoughts) minus that between them (all NHANES respondents: +0.10), with 95% bootstrap intervals for models. Similarity: correlation of the group's 36 residual correlations with those of all NHANES respondents. Real$_=$: real respondents drawn at the model's totals, one per response, mean over 100 draws (2.5th percentile in brackets). No covariation: responses drawn, one per model response, from the maximum-entropy distribution with the model's item distributions at each total, mean over 100 draws (97.5th percentile in brackets). Model + noise: each item moved one level up or down with a probability chosen so that diversity reaches that of real respondents with the same totals, mean over 20 draws. Bottom rows: the profiles given in interviews and the PHQ-9 rated from those interviews (primary rater), pooled over the five patient models, which were given the same profiles; intervals resample profiles, and Real$_=$ draws as many real respondents as there are profiles, at their totals (200 draws).

	Mild		Moderate		Mod. severe		Above, all	
Model	Med.	Above	Med.	Above	Med.	Above	Naive	Coached
GPT-4o-mini	12	100	16	96	19	42	79	58
GPT-4o	12	92	16	78	21	72	81	39
GPT-4.1-nano	13	90	18	100	21	78	89	51
GPT-4.1-mini	12	96	19	92	21	100	96	56
GPT-4.1	9	28	16	96	21	100	75	25
GPT-5-mini	11	88	15	54	21	90	77	53
GPT-5.2	11	72	17	100	21	96	89	59
GPT-5.4-mini	12	96	17	88	22	98	94	53
Haiku 4.5	13	100	19	100	23	96	99	41
Grok-4.1	12	88	16	86	20	88	87	35
Llama-3.3	13	100	15	52	21	86	79	35
Kimi-K2.5	12	98	15	54	18	30	61	31
Mistral-L3	12	100	18	100	22	96	99	39
Phi-4	17	100	21	100	23	100	100	98
Gemini-2.5-FL	13	100	18	100	20	58	86	50
Gemini-2.5-F	12	98	19	96	22	92	95	79
Gemini-2.5-Pro	11	84	16	90	21	84	86	57

Table 38: PHQ-9 totals by instructed severity. Med.: median total; Above: % of responses above the instructed band (mild 5–9, moderate 10–14, moderately severe 15–19), naive arm. Above, all: mean over the three bands, naive and coached arms.

G Additional Figures

Figures 6, 8 and 10 are the full versions of Figures 1, 3 and 4; Figures 7 and 9 show the full feigning comparison and the interventions.

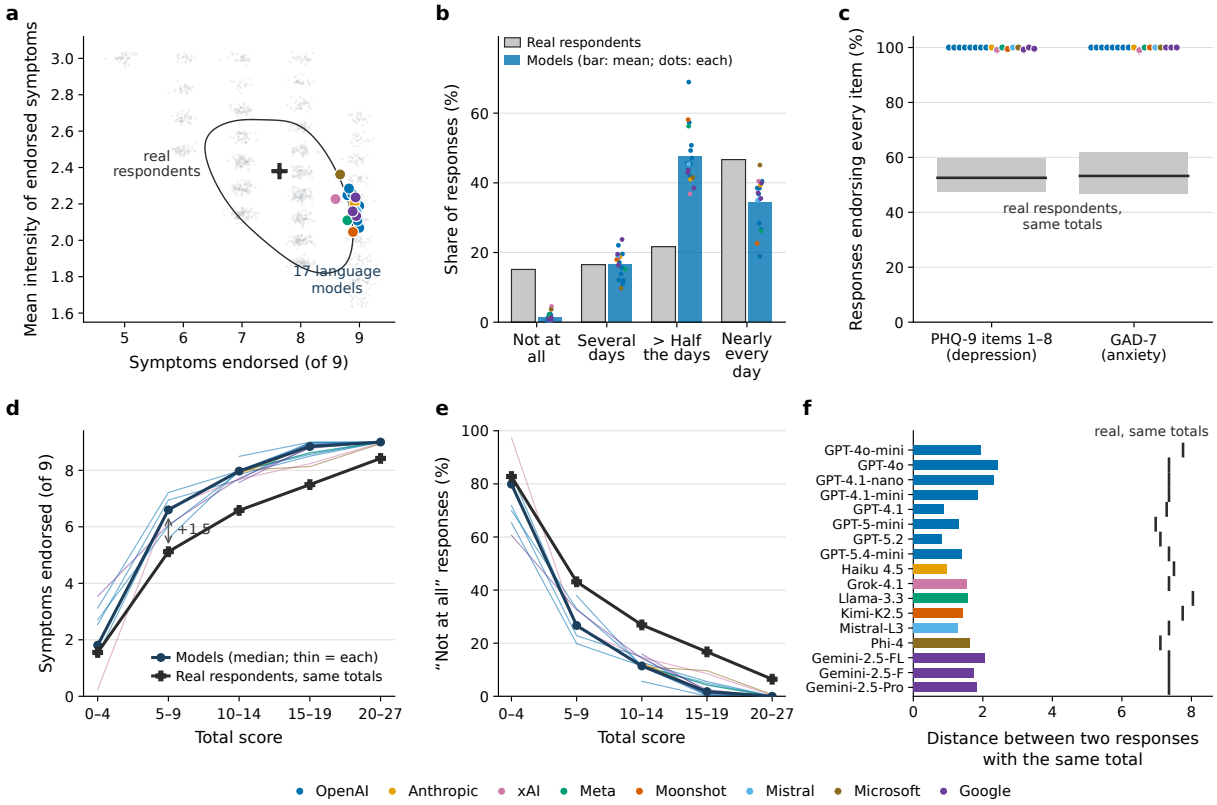


Figure 6: **Response profiles on the PHQ-9 and GAD-7, full version.** Panels a, b and f use PHQ-9 responses with total 15–24. **a**, Symptoms endorsed against their mean intensity (grey dots, real respondents; contour, their densest half; coloured, model means). **b**, Use of the response options. **c**, Responses endorsing every item, PHQ-9 items 1–8 (total 15–24) and GAD-7 (12–19), with 95% intervals; grey, real respondents with the same totals as each model (range and median over models). **d**, **e**, Symptoms endorsed and "not at all" answers by PHQ-9 total, against real respondents with the same totals. **f**, Diversity; ticks, real respondents at each model's totals.

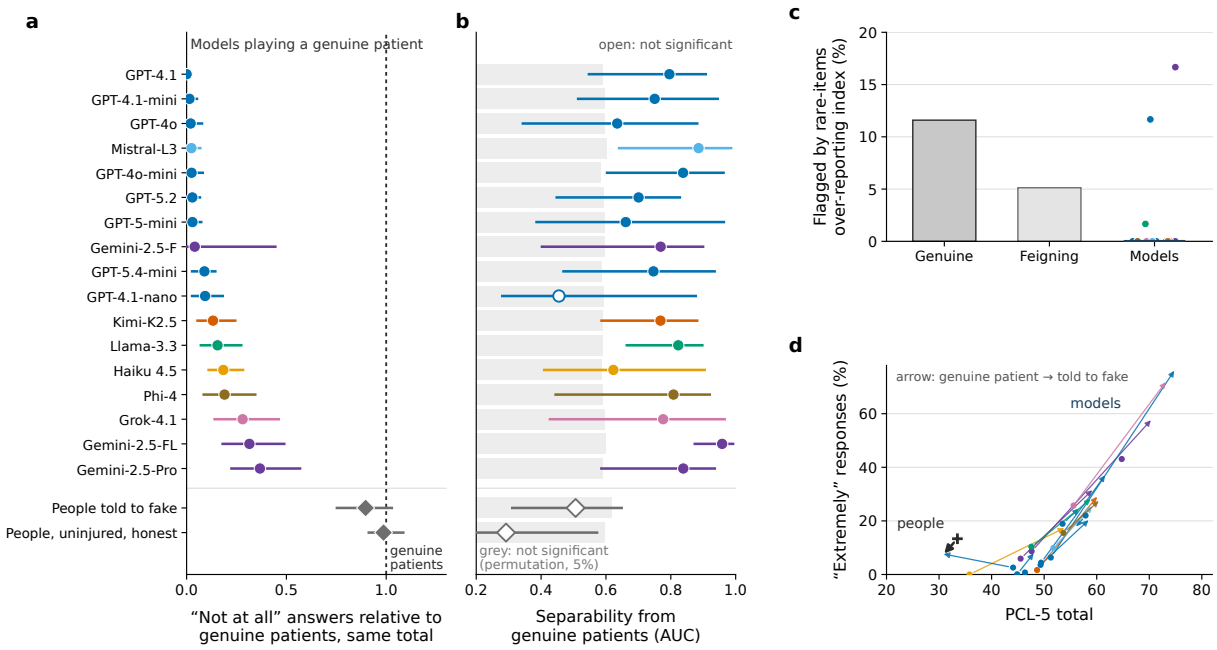


Figure 7: **Comparison with people instructed to feign** (PCL-5), full version. **a**, "Not at all" answers relative to genuinely injured students with the same totals (95% bootstrap intervals). **b**, Separability from them, with the 2.5th–97.5th percentiles over 80% subsamples; grey, below each group's one-sided 5% critical value of a label-permutation test; open, not significant (Holm-adjusted for models). **c**, Rate flagged by the rare-items over-reporting index. **d**, Change in total and in "extremely" answers from the patient to the feigning role.

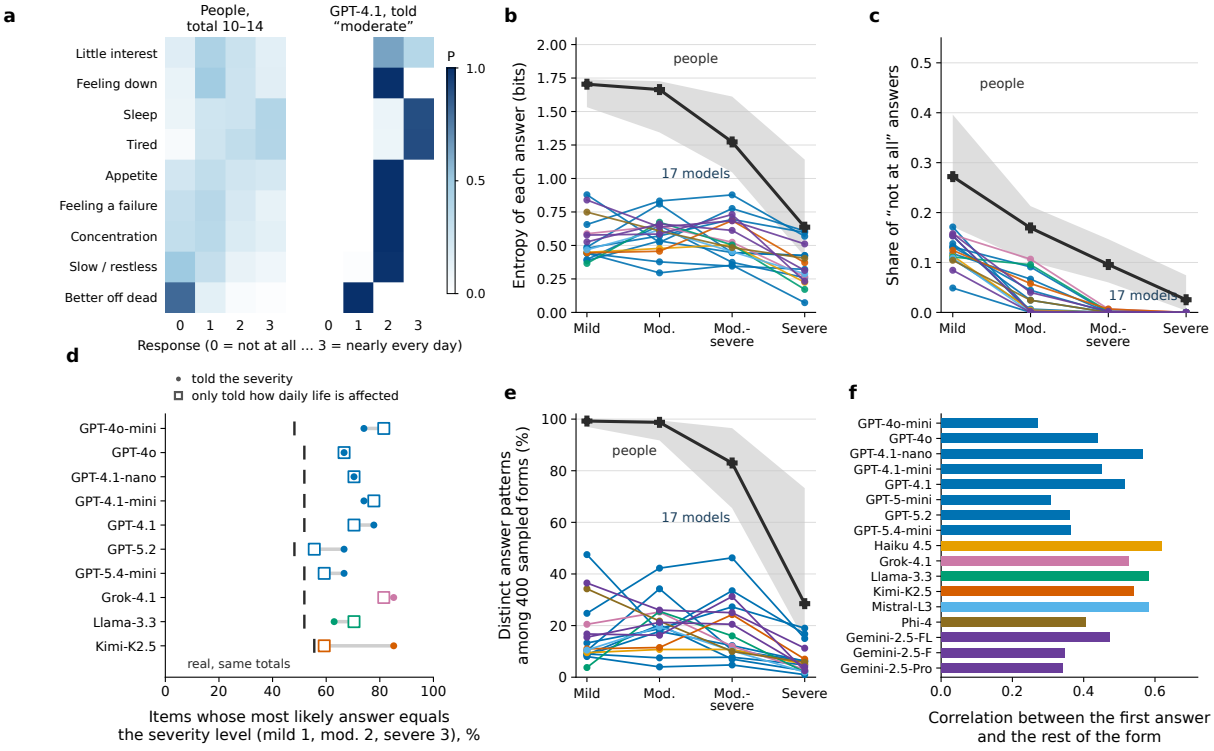


Figure 8: **Per-item answer distributions, full version. a**, Answer probabilities per PHQ-9 item: real respondents with total 10–14 and GPT-4.1 told "moderate" (log-probabilities). **b**, Entropy of each item's answers across the whole-form responses of a severity cell. **c**, Share of "not at all" answers. **d**, Items whose most likely answer, from log-probabilities, equals the severity level, told the severity (dots) or only how daily life is affected (squares); ticks, real respondents with the same totals as the model's whole forms; moderately severe is left out. **e**, Distinct profiles among 400 forms sampled item by item from the answers in b. **f**, Correlation between the first answer and the rest of the form. People in b, c, e: real respondents with the same totals as each model's forms (grey, range over models; line, median).

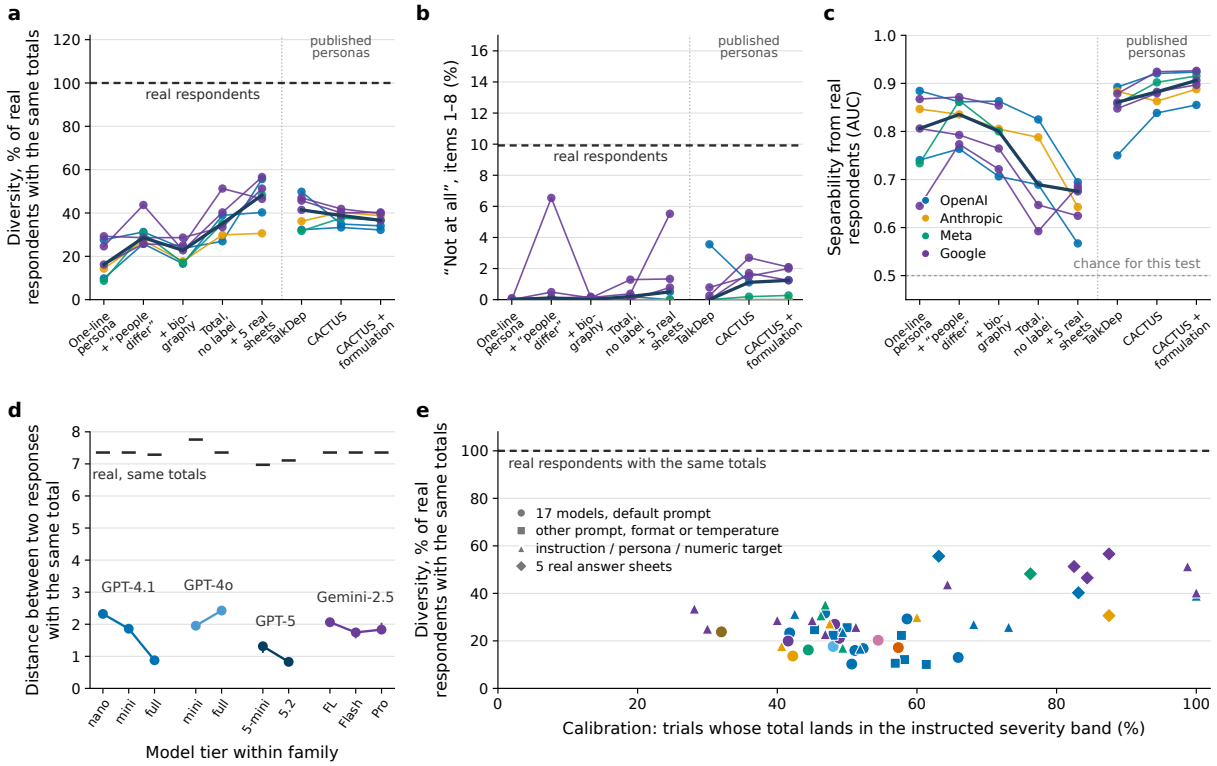


Figure 9: **Interventions, published personas and model scale. a–c**, Diversity as a percentage of held-out real respondents with the same totals (totals 5–27), "not at all" answers and separability (totals 15–24) with the one-line persona, four changes to it, and the TalkDep and CACTUS personas, CACTUS also with each client's case formulation (thin, models; thick, median; dashed, held-out real respondents or chance). **d**, Diversity (totals 15–24) by model tier within four families, with the 2.5th–97.5th percentiles over 80% subsamples; dashes, real respondents at each model's totals. **e**, Calibration against diversity (as in a), all configurations.

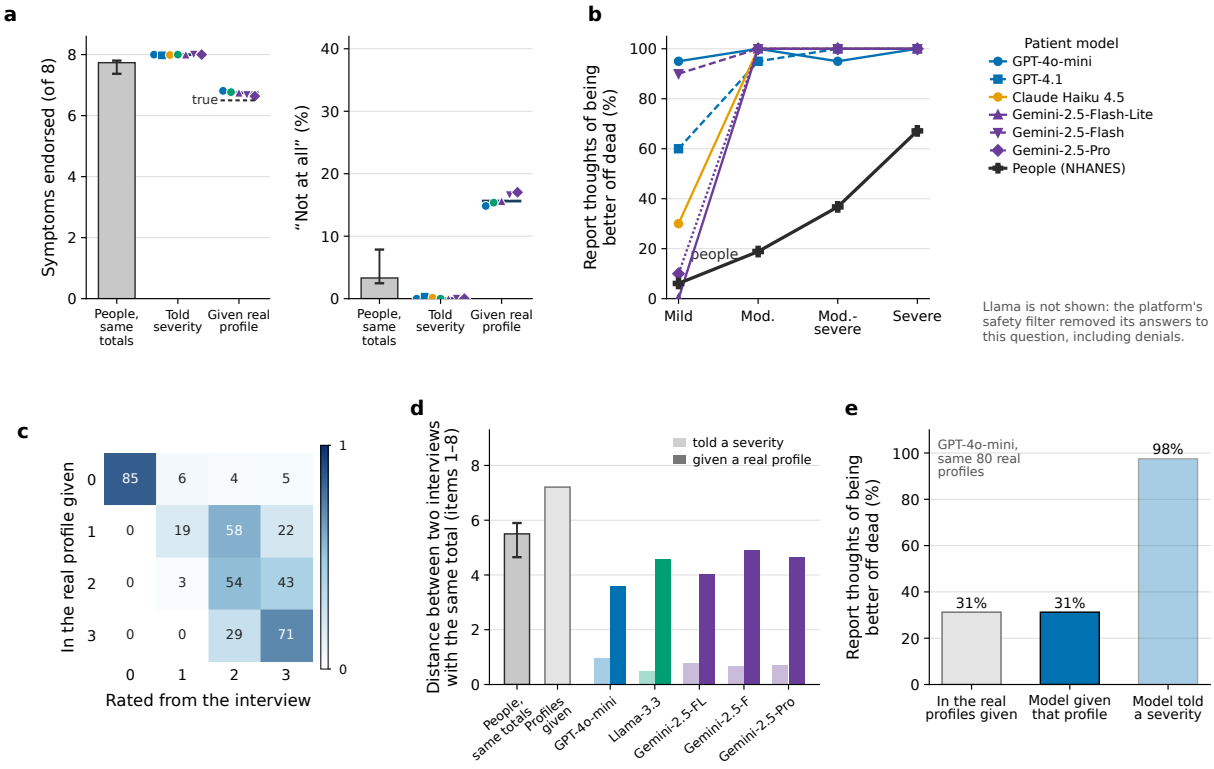


Figure 10: **Simulated clinical interviews, full version. a**, Symptoms endorsed and "not at all" answers, PHQ-9 items 1–8 rated from transcripts (people, real respondents with the same totals; dashed, the profiles given). **b**, Reported thoughts of being better off dead by severity. **c**, Given against rated answer, row percentages. **d**, Diversity. **e**, Suicidal ideation in the real profiles given, in the interviews built on them, and with a severity persona (GPT-4o-mini).

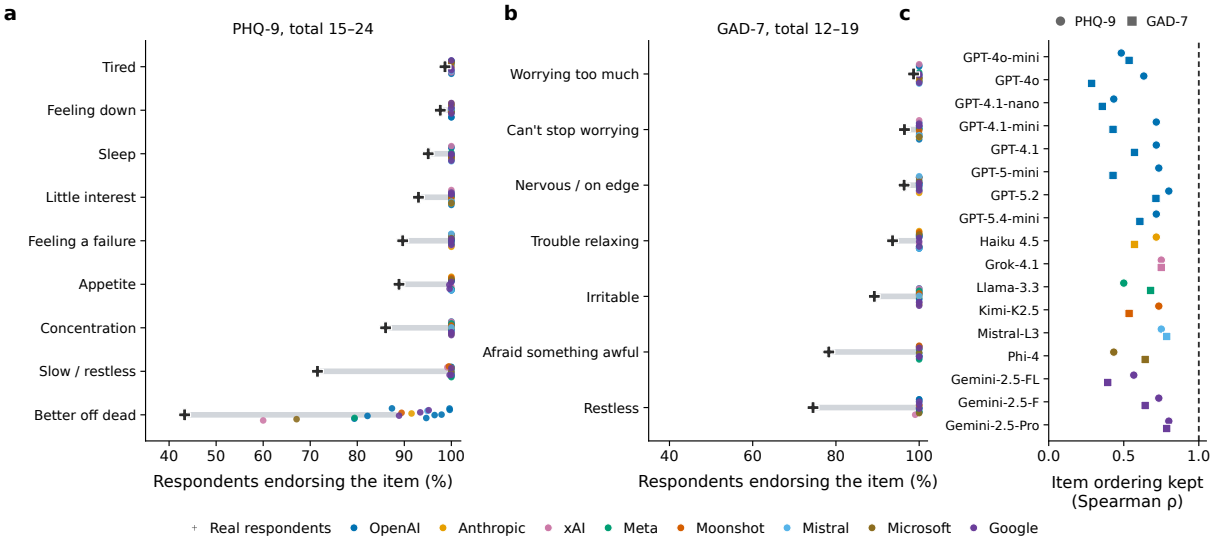


Figure 11: **Item endorsement on the PHQ-9 and GAD-7. a**, **b**, Share endorsing each item, PHQ-9 total 15–24 and GAD-7 total 12–19; crosses, real respondents. **c**, Spearman correlation with the human item ordering.

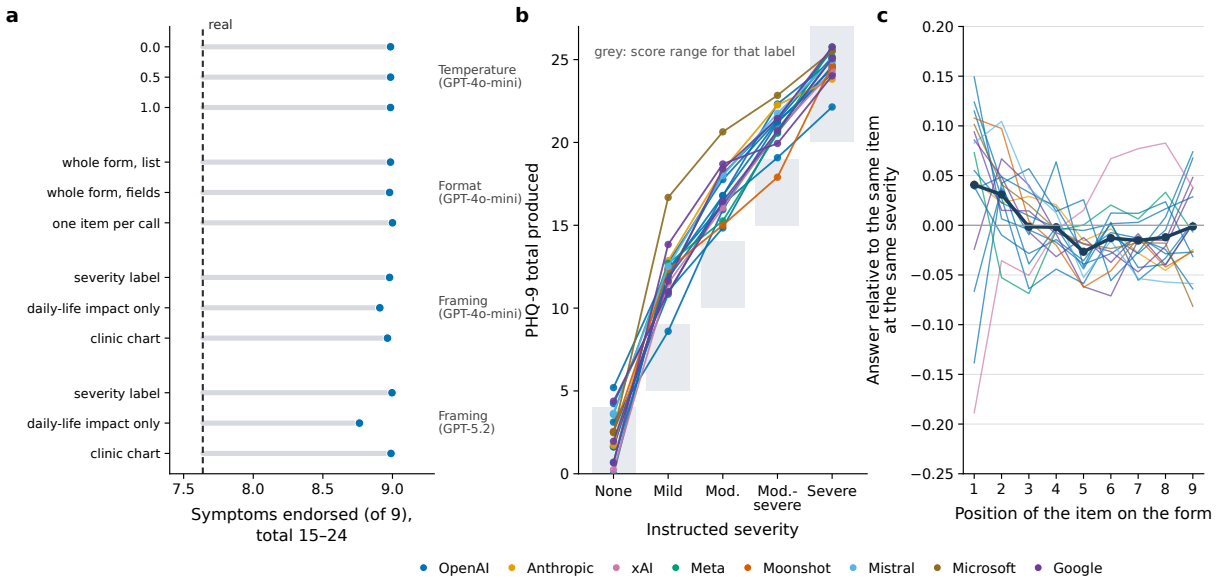


Figure 12: **Robustness. a**, Symptoms endorsed (PHQ-9 total 15–24) under different temperatures, answer formats and persona framings. **b**, Total produced against instructed severity. **c**, Answer by position on the form, relative to the same item at the same severity.

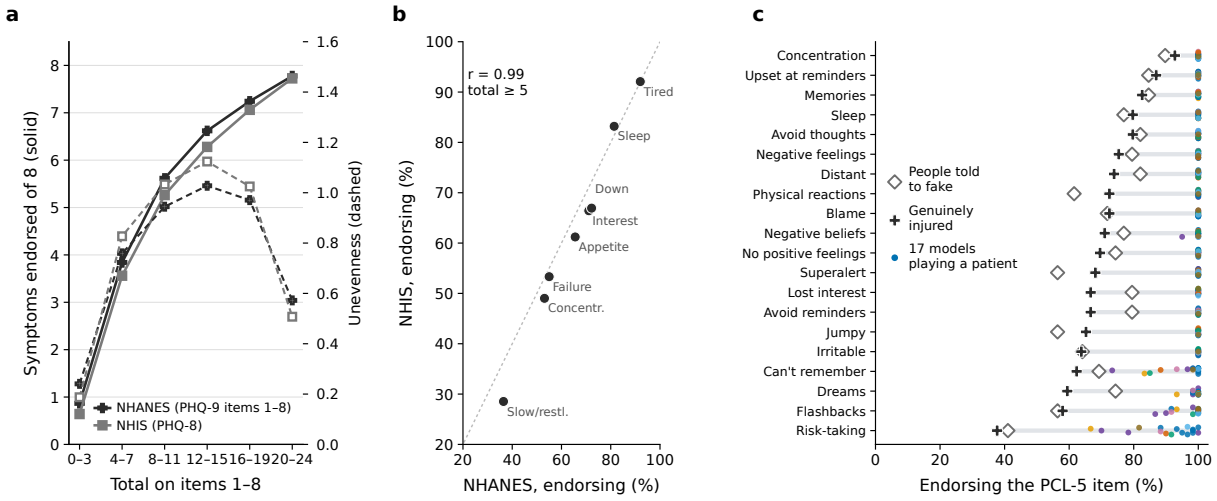


Figure 13: **Checks on the human references. a**, Symptoms endorsed and unevenness by score band, NHANES and NHIS. **b**, Item endorsement in the two surveys, total ≥5. **c**, PCL-5 item endorsement in the feigning study and in the models.

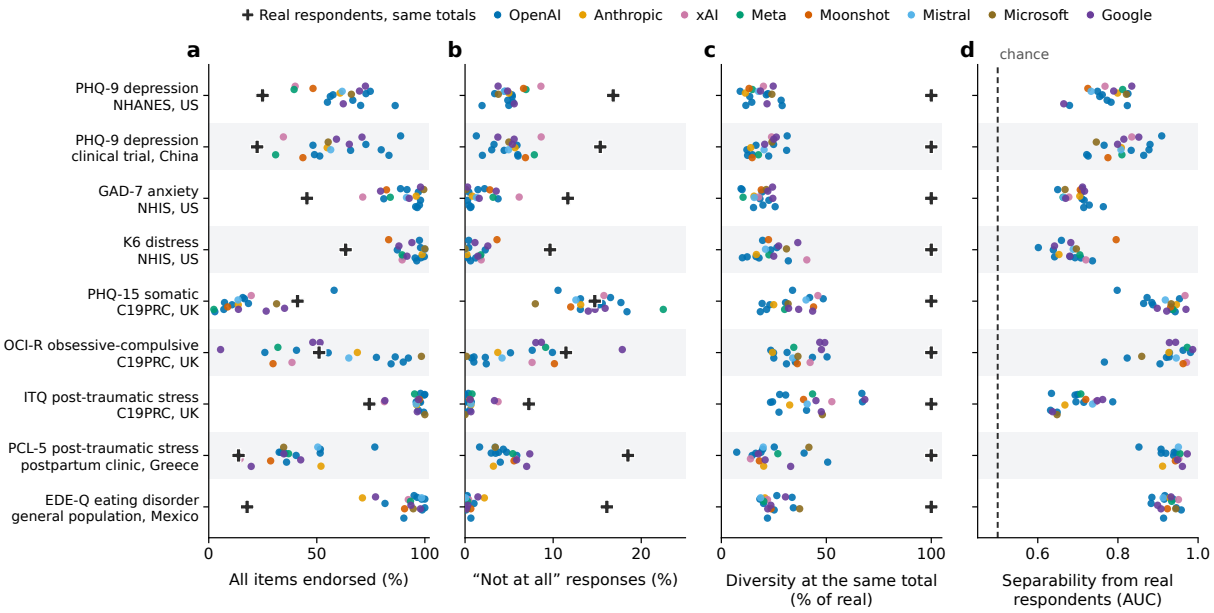


Figure 14: **Profile measures on the nine human references.** **a**, Responses endorsing every item. **b**, "Not at all" answers. **c**, Diversity as a percentage of the real value. **d**, Separability; dashed line, chance. Crosses, real respondents with the same totals.

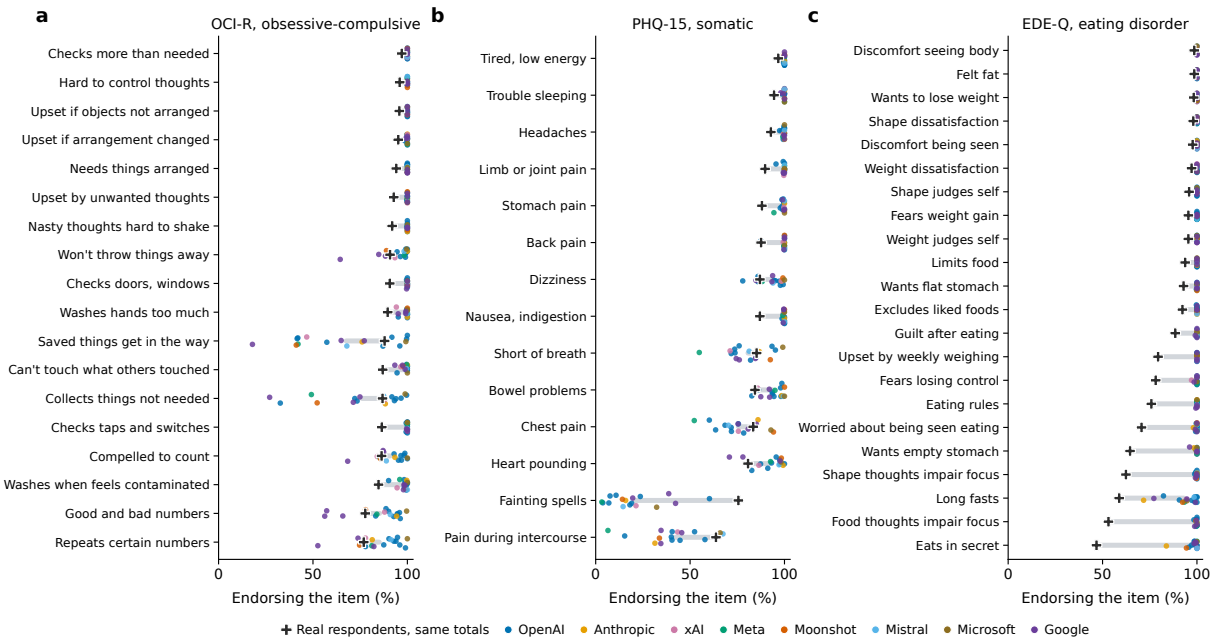


Figure 15: **Item endorsement at matched totals.** **a**, OCI-R. **b**, PHQ-15. **c**, EDE-Q. Crosses, real respondents with the same totals.

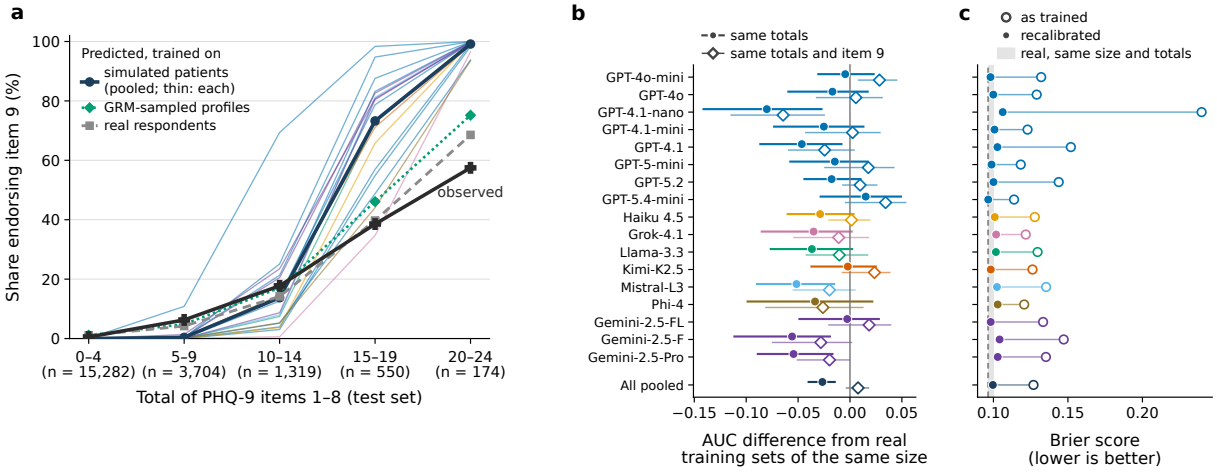


Figure 16: **Screening model trained on simulated patients.** Predicting whether PHQ-9 item 9 is endorsed from items 1–8; trained on NHANES 2005–2012 or on simulated patients, tested on NHANES 2013–2023. **a**, Predicted and observed rates by total (thin, per model). **b**, AUC difference from real training sets of the same size with the same items 1–8 totals (dots) and also the same item-9 answers (diamonds); 95% paired bootstrap intervals. **c**, Brier score as trained and after recalibration on held-out real respondents; grey, real training sets of the same size and totals; dashed, the full real training set.